Online Supplementary Materials for "A General Framework for Incorporating Identification Uncertainty in Individualized Treatment Rules" by Liang, Ye, and Zhao

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Abstract

This document include supplementary materials for the paper entitled "A General Framework for Incorporating Identification Uncertainty in Individualized Treatment Rules" by Liang, Ye, and Zhao. It includes details on the implementation of the proposed method, its theoretical properties and their proofs, and additional simulation and real data analysis using the proposed methods.

Keywords: Individualized treatment rules, Identification uncertainty, Instrumental variable, Partial identification, Deferred option

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1 Implementation

In this section, we present the algorithm for solving the augmented empirical risk minimization (ERM) problem and its integration with the cross-fitting algorithm. First, given the estimators $\widehat{p}_{y,a|z}(\boldsymbol{X})$ and $\widehat{\pi}_z(\boldsymbol{X})$, we discuss how to solve

$$E_n\left[\left\{\widehat{U}_+(\boldsymbol{X}) - B_+(\boldsymbol{X},Y,A,Z)\right\}\phi_d(f(\boldsymbol{X})) + \left\{\widehat{L}_-(\boldsymbol{X}) - B_-(\boldsymbol{X},Y,A,Z)\right\}\phi_d(-f(\boldsymbol{X}))\right] + \lambda_n \|f\|_{\mathcal{H}}^2, \quad (1)$$

where $B_+(\boldsymbol{X},Y,A,Z)$ and $B_-(\boldsymbol{X},Y,A,Z)$ are estimates of the biases of $\widehat{L}(\boldsymbol{X})$ and $\widehat{U}(\boldsymbol{X})$ w.r.t $L(\boldsymbol{X})$ and $U(\boldsymbol{X})$. Denote

$$\begin{aligned} W_+(\boldsymbol{X},Y,A,Z) &= \widehat{U}_+(\boldsymbol{X}) - B_+(\boldsymbol{X},Y,A,Z), \\ W_-(\boldsymbol{X},Y,A,Z) &= \widehat{L}_-(\boldsymbol{X}) - B_-(\boldsymbol{X},Y,A,Z). \end{aligned}$$

Consider an RKHS $\mathcal{H}$ with a kernel function $K(\cdot,x)$, where $\langle K(\cdot,x), f\rangle = f(\boldsymbol{X})$ holds for any $f \in \mathcal{H}$, and $\langle\cdot,\cdot\rangle$ represents the inner product equipped by $\mathcal{H}$. By utilizing the representer lemma, the minimization of (1) can be written as the minimization of

$$\begin{aligned} &E_n\left[W_+(\boldsymbol{X},Y,A,Z)\phi_d\left\{\sum_{j=1}^n \alpha_j K(X,X_j)\right\} + W_-(\boldsymbol{X},Y,A,Z)\phi_d\left\{-\sum_{j=1}^n \alpha_j K(X,X_j)\right\}\right] \\ &+\lambda_n \sum_{i,j=1}^n \alpha_i\alpha_j K(X_i,X_j), \end{aligned}$$

w.r.t. $\boldsymbol{\alpha} = (\alpha_1,\cdots,\alpha_n)$. Unlike the traditional empirical risk minimization where the objective function is convex w.r.t $\boldsymbol{\alpha}$, the objective in (1) may not be convex. However, the objective function can be written into the difference of two convex functions, i.e.,

$$\left\{\ell_n^+(\boldsymbol{\alpha}) + \lambda_n \sum_{i,j=1}^n \alpha_i\alpha_j K(X_i,X_j)\right\} - \ell_n^-(\boldsymbol{\alpha}),$$

where

$$\begin{aligned}
\ell_n^+(\boldsymbol{\alpha}) &= E_n\left[\{W_+(\boldsymbol{X},Y,A,Z)\}_+\,\phi_d\left\{\sum_{j=1}^n \alpha_j K(X,X_j)\right\}\right.\\
&\quad\left.+\{W_-(\boldsymbol{X},Y,A,Z)\}_+\,\phi_d\left\{-\sum_{j=1}^n \alpha_j K(X,X_j)\right\}\right],\\
\ell_n^-(\boldsymbol{\alpha}) &= E_n\left[\{W_+(\boldsymbol{X},Y,A,Z)\}_-\,\phi_d\left\{\sum_{j=1}^n \alpha_j K(X,X_j)\right\}\right.\\
&\quad\left.+\{W_-(\boldsymbol{X},Y,A,Z)\}_-\,\phi_d\left\{-\sum_{j=1}^n \alpha_j K(X,X_j)\right\}\right].
\end{aligned}$$

Notice that $\ell_n^+(\boldsymbol{\alpha}) + \lambda_n \sum_{i,j=1}^n \alpha_i\alpha_j K(X_i,X_j)$ is strictly convex w.r.t $\boldsymbol{\alpha}$, and $\ell_n^-(\boldsymbol{\alpha})$ is convex w.r.t $\boldsymbol{\alpha}$. Thus, the proposed objective can be decomposed as the difference of a strictly convex function and a convex function with respect to $\boldsymbol{\alpha}$. Consequently, we employ the difference of convex algorithm (DCA) introduced in Tao and An (1997). In DCA, we need to specify initial values for $\boldsymbol{\alpha}$, which can be obtained from the traditional empirical risk minimization approach (without augmentations).

Now, we present the complete algorithm incorporating cross-fitting. First, we split the entire data into two folds with equal sample sizes. Then we use the samples in the first fold to estimate $p_{y,a|z}(\boldsymbol{X})$'s and $\pi_z(\boldsymbol{X})$'s using non-parametric or machine learning methods. After obtaining the $\widehat{p}_{y,a|z}(\boldsymbol{X})$'s and $\widehat{\pi}_z(\boldsymbol{X})$'s, using the samples on the second fold, we construct the weights $W_+(\boldsymbol{X},Y,A,Z)$ and $W_-(\boldsymbol{X},Y,A,Z)$, and then solve the empirical augmented risk minimization problem using the DCA. Denote the resulting minimizer as $\widehat{f}_d^{(1)}$. Following the cross-fitting algorithm, we estimate $p_{y,a|z}(\boldsymbol{X})$'s and $\pi_z(\boldsymbol{X})$'s using the samples in the second fold and construct the weights, and then solve the empirical augmented risk minimization problem using the samples in the first fold. This yields another minimizer, denoted as $\widehat{f}_d^{(2)}$. Finally, we obtain the final estimator as $\widehat{f}_d(\boldsymbol{X}) = \frac{\widehat{f}_d^{(1)}(\boldsymbol{X})+\widehat{f}_d^{(2)}(\boldsymbol{X})}{2}$. To convert the estimator $\widehat{f}_d(\boldsymbol{X})$ into an ITR with a deferred option, we employ $D(\boldsymbol{X};\widehat{f}_d,\delta)$ with a pre-specified δ. This cross-fitting procedure can be extended to allow more than two folds.

2 Extension to continuous outcome

In this section, we extend our proposed method to deal with continuous outcome. Without the loss of generality, we assume that $Y \in [0,1]$. We adopt the idea in Levis et al. (2025), where an independent uniform random variables is introduced to provide a partial identification bound. Specifically, let V be a random variable following a uniform distribution on $[0,1]$ and independent from the observed data $(\boldsymbol{X}, Y, A, Z)$. Following the idea of Section 6 in Levis et al. (2025), we have the following theorem.

Theorem 1. *Under the core IV conditions, let* $Y(v) = 1\{Y > v\}$ *(similar binary potential outcomes can be defined), suppose that for any fixed* v*, we have* $L(\boldsymbol{X}; v) \leqslant E[Y_1(v) - Y_{-1}(v) \mid \boldsymbol{X}] \leqslant U(\boldsymbol{X}; v)$*, where* $L(\boldsymbol{X}; v) = \max\{P_1(\boldsymbol{X}; v), \cdots, P_K(\boldsymbol{X}; v)\}$ *and* $U(\boldsymbol{X}; v) = \min\{Q_1(\boldsymbol{X}; v), \cdots, Q_K(\boldsymbol{X}; v)\}$*. Then, we have* $L(\boldsymbol{X}) \leqslant \Delta(\boldsymbol{X}) \leqslant U(\boldsymbol{X})$*, where* $L(\boldsymbol{X}) = \max\{E_V[P_1(\boldsymbol{X}; V)], \cdots, E_V[P_K(\boldsymbol{X}; V)]\}$ *and* $U(\boldsymbol{X}) = \min\{E_V[Q_1(\boldsymbol{X}; V)], \cdots, E_V[Q_K(\boldsymbol{X}; V)]\}$.

The proof of Theorem 1 is immediate using the concavity of $\max(\cdot)$ and convexity of $\min(\cdot)$. Theorem 1 shows that for continuous outcomes, using a partial identification bound for $E[Y_1(v) - Y_{-1}(v) \mid \boldsymbol{X}]$, we can derive a partial identification bound for $\Delta(\boldsymbol{X})$. In addition, the partial identification bound for $\Delta(\boldsymbol{X})$ follows the same format as those for binary outcomes in the main text. Thus, our proposed estimation methods can be applied to estimate the EIV-ITR under the continuous outcomes. If we adopt the Balke-Pearl bound for the partial identification of $E[Y_1(v) - Y_{-1}(v) \mid \boldsymbol{X}]$, then $P_k(\boldsymbol{X}; v)$'s and $Q_k(\boldsymbol{X}; v)$'s are linear transformations of $P_{y(v)=1,a|z}(\boldsymbol{X}) = P(Y(v) = 1, A = a \mid Z = z, \boldsymbol{X})$ or $P_{y(v)=0,a|z}(\boldsymbol{X}) = P(Y(v) = 0, A = a \mid Z = z, \boldsymbol{X})$. Following Theorem 1, then the $E_V[P_k(\boldsymbol{X}; V)]$'s and $E_V[Q_k(\boldsymbol{X}; V)]$'s are linear transformations of $P_{1,a|z}(\boldsymbol{X}) = P(Y > V, A = a \mid Z = z, \boldsymbol{X})$ or $P_{0,a|z}(\boldsymbol{X}) = P(Y \leqslant V, A = a \mid Z = z, \boldsymbol{X})$. This implies that we can construct $\widetilde{Y} := 1\{Y > V\}$, where V is an independently generated random variable following a uniform distribution on $[0,1]$, and then use the proposed algorithm with $\widetilde{Y}$ as the observed outcome to estimate the EIV-ITR under the continuous outcomes. Here, for the completeness of the discussion, we provide the explicit formulas

of the Balke-Pearl bound for binary outcomes:

$$L(\boldsymbol{x}) := \max_{k\{1,\cdots,8\}} \{P_k(\boldsymbol{x})\} \leqslant \Delta(\boldsymbol{x}) \leqslant \min_{k\{1,\cdots,8\}} \{Q_k(\boldsymbol{x})\} =: U(\boldsymbol{x}),$$

where

$$P_1(x) = p_{1,1|1}(x) + p_{0,0|0}(x) - 1,$$

$$P_2(x) = p_{1,1|0}(x) + p_{0,0|1}(x) - 1,$$

$$P_3(x) = -p_{0,1|1}(x) - p_{1,0|1}(x),$$

$$P_4(x) = -p_{0,1|0}(x) - p_{1,0|0}(x),$$

$$P_5(x) = p_{1,1|0}(x) - p_{1,1|1}(x) - p_{1,0|1}(x) - p_{0,1|0}(x) - p_{1,0|0}(x),$$

$$P_6(x) = p_{1,1|1}(x) - p_{1,1|0}(x) - p_{1,0|0}(x) - p_{0,1|1}(x) - p_{1,0|1}(x),$$

$$P_7(x) = p_{0,0|1}(x) - p_{0,1|1}(x) - p_{1,0|1}(x) - p_{0,1|0}(x) - p_{0,0|0}(x),$$

$$P_8(x) = p_{0,0|0}(x) - p_{0,1|0}(x) - p_{1,0|0}(x) - p_{0,1|1}(x) - p_{0,0|1}(x).$$

and

$$
\begin{aligned}
Q_1(x) &= 1 - p_{0,1|1}(x) - p_{1,0|0}(x), \\
Q_2(x) &= 1 - p_{0,1|0}(x) - p_{1,0|1}(x), \\
Q_3(x) &= p_{1,1|1}(x) + p_{0,0|1}(x), \\
Q_4(x) &= p_{1,1|0}(x) + p_{0,0|0}(x), \\
Q_5(x) &= -p_{0,1|0}(x) + p_{0,1|1}(x) + p_{0,0|1}(x) + p_{1,1|0}(x) + p_{0,0|0}(x), \\
Q_6(x) &= -p_{0,1|1}(x) + p_{1,1|1}(x) + p_{0,0|1}(x) + p_{0,1|0}(x) + p_{0,0|0}(x), \\
Q_7(x) &= -p_{1,0|1}(x) + p_{1,1|1}(x) + p_{0,0|1}(x) + p_{1,1|0}(x) + p_{1,0|0}(x), \\
Q_8(x) &= -p_{1,0|0}(x) + p_{1,1|0}(x) + p_{0,0|0}(x) + p_{1,1|1}(x) + p_{1,0|1}(x).
\end{aligned}
$$

In the following, we provide a simulation following the Setting (1,2) with potential outcomes $Y(a)$'s following:

$$
Y_a = \text{logit}\left\{0.5g_1(\boldsymbol{X}, U) + 0.5g_2(\boldsymbol{X}, U, a) + 0.1\epsilon + 1\right\},
$$

where ϵ is generated from a standard normal distribution, and $\text{logit}(t) = \exp(t)/\{1 + \exp(t)\}$. Implementation of the proposed methods and comparison methods follow the description in the main text. Figure 1 shows the results when the outcome is continuous. We can see that the proposed augmented ERM yields lower loss in value and higher value for non-deferred patients with minimal deferred proportion across all choices of d and ξ.

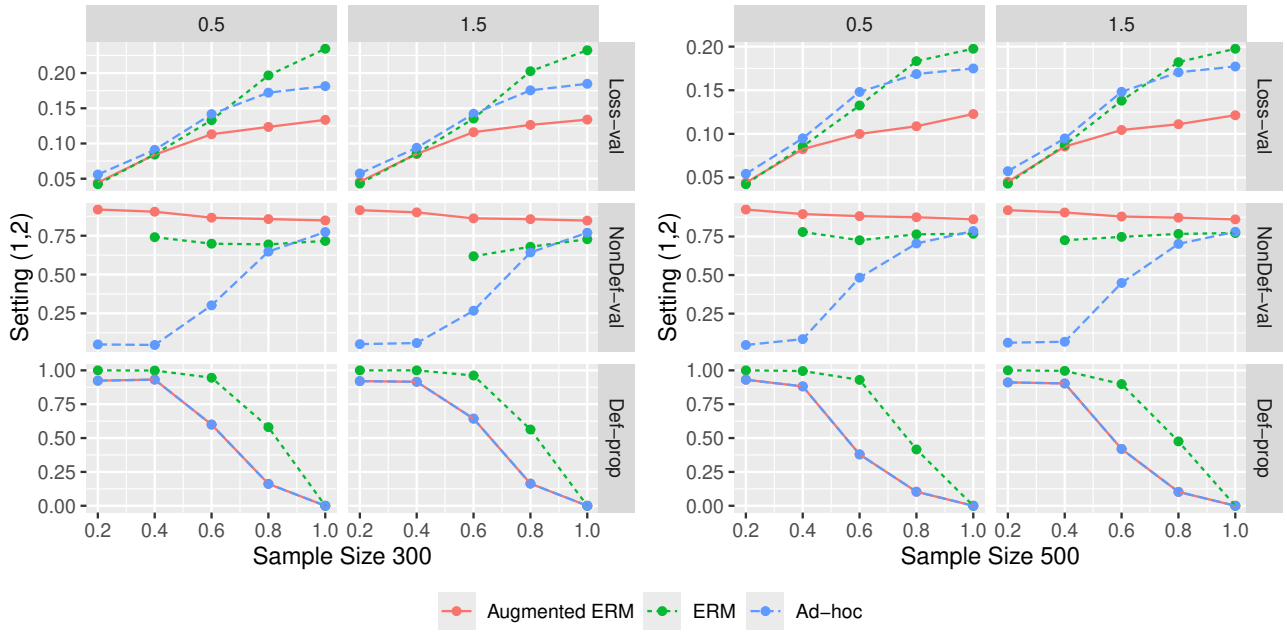


Figure 1: Simulation results for continuous outcomes over 200 repeats.

3 Proofs of Theorems

3.1 Excess risk bounds

In this section, we provide the excess risk bound for $R\{D(\boldsymbol{X}; f_d, \delta)\} - R^*$. The excess risk bound aims to bound $R\{D(\boldsymbol{X}; f_d, \delta)\} - R^*$ using the excess risk of the surrogate loss, i.e. $R_{\phi_d}(f) - \inf_f R_{\phi_d}(f)$.

Theorem 2. *For all* $0 < d \leqslant 1$ *and* $0 < \delta < 1$, *we have the following:*

$$R(D) - R^* \leqslant \max\left\{\frac{1}{1-\delta}, \frac{d}{\delta(1+d)}\right\}\left\{R_{\phi_d}(f) - R^*_{\phi_d}\right\};$$

For $\delta = 0$ *and* $d = 1$, *we have*

$$R(D) - R^* \leqslant R_{\phi_d}(f) - R^*_{\phi_d}.$$

Proof. The entire proof consists of three parts. In Part I, we derive the Bayes decision rules that

minimize $R(D)$ and $R_{\phi_d}(f)$, where

$$R(D) = E\left[\sup_{\Delta(\boldsymbol{X})\in[L(\boldsymbol{X}),U(\boldsymbol{X})]} \widetilde{TLV}(D,\boldsymbol{X};\Delta,d)\right].$$

and

$$R_{\phi_d}(f) = E\left[U_+(\boldsymbol{X})\phi_d(f(\boldsymbol{X})) + L_-(\boldsymbol{X})\phi_d(-f(\boldsymbol{X}))\right].$$

In Part II, we derive an intermediate result, which can be used to prove the excess risk bound. In Part III, we use the result derived in Part I and II to prove the excess risk bound.

We will first derive the Bayes decision rules that minimize $R(D)$ and $R_{\phi_d}(f)$. For simplicity of the notations, we drop the X in $\Delta(\boldsymbol{X})$, $L(\boldsymbol{X})$, and $U(\boldsymbol{X})$, and use Δ, L, U instead. We define

$$\begin{aligned} R_X(D) &= \sup_{\Delta\in[L,U]} \widetilde{TLV}(D,\boldsymbol{X};\Delta,d), \\ R_{X,\phi_d}(f) &= U_+(\boldsymbol{X})\phi_d(f(\boldsymbol{X})) + L_-(\boldsymbol{X})\phi_d(-f(\boldsymbol{X})). \end{aligned}$$

Notice that $R(D) = E[R_X(D)]$ and $R_{\phi_d}(f) = E[R_{X,\phi_d}(f)]$. Thus, to derive the Bayes decision rules that minimize $R(D)$ and $R_{\phi_d}(f)$, it suffices to derive the minimizers for $R_X(z)$ for $z \in \{-1, 0, 1\}$ and $R_{X,\phi_d}(z)$ for $z \in \mathbb{R}$. Notice that

$$\begin{aligned} R_X(z) &= \sup_{\Delta\in[L,U]} \widetilde{TLV}(z,\boldsymbol{X};\Delta,d) \\ &= \sup_{\Delta\in[L,U]} \left\{|\Delta| 1\left\{z \neq \operatorname{sgn}(\Delta)\right\} 1\left\{z \neq 0\right\} + d|\Delta| 1\left\{z = 0\right\}\right\}. \end{aligned}$$

Thus, we have

$$\begin{aligned} R_X(1) &= \sup_{\Delta\in[L,U]} |\Delta| 1\{1 \neq \operatorname{sgn}(\Delta)\} = L_-, \\ R_X(-1) &= \sup_{\Delta\in[L,U]} |\Delta| 1\{-1 \neq \operatorname{sgn}(\Delta)\} = U_+, \\ R_X(0) &= \sup_{\Delta\in[L,U]} d|\Delta| = d\max\{U_+, L_-\}. \end{aligned}$$

By direct calculation, we can know that the minimizer of $R_X(z)$ over $z \in \{-1, 0, 1\}$ is

$$z^* = \begin{cases} 1 & \text{if } L_-/U_+ < d, \\ -1 & \text{if } U_+/L_- < d, \\ 0 & \text{otherwise.} \end{cases}$$

Thus, the D_d^* minimizes the $R(D)$. In addition, the $R^* = R(D_d^*) = E[R_X(z^*)]$, where

$$R_X(z^*) = \begin{cases} 0 & \text{if } LU \geqslant 0, \\ \min\{L_-, U_+, d\max\{L_-, U_+\}\} & \text{if } LU < 0. \end{cases}$$

Next, we minimize $R_{X,\phi_d}(z)$ over $z \in \mathbb{R}$. By direct calculation, we have

$$R_{X,\phi_d}(z) = \begin{cases} U_+(1 - z/d) & \text{if } z < -1, \\ U_+(1 - z/d) + L_-(1 + z) & \text{if } -1 \leqslant z < 0, \\ U_+(1 - z) + L_-(1 + z/d) & \text{if } 0 \leqslant z \leqslant 1, \\ L_-(1 + z/d) & \text{if } z > 1. \end{cases}$$

It can be verified that z^* is the minimizer of $R_{X,\phi_d}(z)$, and

$$R_{X,\phi_d}(z^*) = \begin{cases} 0 & \text{if } LU \geqslant 0, \\ U_+ + L_- + \min\left\{0, U_+/d - L_-, L_-/d - U_+\right\} & \text{if } LU < 0. \end{cases}$$

This concludes the Part I of the proof.

In Part II of the proof, we will show the following lemma. Define

$$\begin{aligned} R_{-1}(\boldsymbol{X}) &= \inf_{z \leqslant -\delta} R_{X,\phi_d}(z) \\ &= \min\left\{U_+(1+1/d), U_+(1+\delta/d) + L_-(1-\delta)\right\}, \\ R_{+1}(\boldsymbol{X}) &= \inf_{z \geqslant \delta} R_{X,\phi_d}(z) \\ &= \min\left\{L_-(1+1/d), L_-(1+\delta/d) + U_+(1-\delta)\right\}, \\ R_0(\boldsymbol{X}) &= \inf_{|z| \leqslant \delta} R_{X,\phi_d}(z) \\ &= \min\left\{U_+(1+\delta/d) + L_-(1-\delta), L_-(1+\delta/d) + U_+(1-\delta), U_+ + L_-\right\}. \end{aligned}$$

Lemma 1. *We have the following:*

1. *When $LU \geqslant 0$, we have*

$$R_X(z) - R_X(z^*) \leqslant \frac{1}{1-\delta}\left\{R_z(\boldsymbol{X}) - R_{X,\phi_d}(z^*)\right\}.$$

2. *When $LU < 0$, and $0 < \delta < 1$, we have*

$$R_X(z) - R_X(z^*) \leqslant \max\left\{\frac{d}{1-\delta}, \frac{d}{\delta(1+d)}\right\}\left\{R_z(\boldsymbol{X}) - R_{X,\phi_d}(z^*)\right\};$$

For $\delta = 0$ and $d = 1$, we have

$$R_X(z) - R_X(z^*) \leqslant R_z(\boldsymbol{X}) - R_{X,\phi_d}(z^*).$$

To show this lemma, we first consider the scenario where $LU \geqslant 0$. When $LU \geqslant 0$, we have

$$\begin{aligned} R_X(z) - R_X(z^*) &= L_- 1\{z = 1\} + U_+ 1\{z = -1\} \\ &\quad + d \max\{L_-, U_+\} 1\{z = 0\}, \end{aligned}$$

and $R_z(\boldsymbol{X}) - R_{X,\phi_d}(z^*) = R_z(\boldsymbol{X})$.

We will verify

$$R_X(z) - R_X(z^*) \leqslant \frac{1}{1-\delta} \{R_z(\boldsymbol{X}) - R_{X,\phi_d}(z^*)\}.$$

Specifically, it suffices to verify that

$$\begin{aligned}
R_X(1) - R_X(z^*) &= L_- \\
&\leqslant \frac{1}{1-\delta} R_{+1}(\boldsymbol{X}) \\
&= \frac{1}{1-\delta} \min\left\{L_-(1+1/d), L_-(1+\delta/d) + U_+(1-\delta)\right\}; \\
R_X(-1) - R_X(z^*) &= U_+ \\
&\leqslant \frac{1}{1-\delta} R_{-1}(\boldsymbol{X}) \\
&= \frac{1}{1-\delta} \min\left\{U_+(1+1/d), U_+(1+\delta/d) + L_-(1-\delta)\right\}; \\
R_X(0) - R_X(z^*) &= d \max\left\{L_-, U_+\right\} \\
&\leqslant \frac{1}{1-\delta} R_0(\boldsymbol{X}) \\
&= \frac{1}{1-\delta} \min\left\{U_+(1+\delta/d) + L_-(1-\delta),\right. \\
&\quad \left. L_-(1+\delta/d) + U_+(1-\delta), U_+ + L_-\right\}.
\end{aligned}$$

To verify these inequalities, notice that when $LU \geqslant 0$, either L_- or U_+ is 0. When $L_- = 0$, these

inequalities are equivalent to

$$\begin{aligned}
R_X(1) - R_X(z^*) &= 0 \\
&\leqslant \frac{1}{1-\delta} R_{+1}(\boldsymbol{X}) \\
&= 0; \\
R_X(-1) - R_X(z^*) &= U_+ \\
&\leqslant \frac{1}{1-\delta} R_{-1}(\boldsymbol{X}) \\
&\leqslant \frac{1}{1-\delta} U_+(1+\delta/d); \\
R_X(0) - R_X(z^*) &= dU_+ \\
&\leqslant \frac{1}{1-\delta} R_0(\boldsymbol{X}) \\
&= \frac{1}{1-\delta} U_+(1-\delta).
\end{aligned}$$

These inequalities can be easily verified via direct calculation. Similar results hold when $U_+ = 0$.

When $LU \leqslant 0$, we have that both L_- and U_+ are non-zeros. Without loss of generality, we assume that $L_- \leqslant U_+$. In this case, we have

$$\begin{aligned}
R_X(z) - R_X(z^*) &= \max\{0, L_- - dU_+\} 1\{z=1\} + \max\{U_+ - L_-, (1-d)U_+\} 1\{z=-1\} \\
&\quad + \max\{0, dU_+ - L_-\} 1\{z=0\}.
\end{aligned}$$

and $R_{X,\phi_d}(z^*) = \min\{U_+ + L_-, (1+1/d)L_-\}$. We separate our discussion into two scenarios. When $dU_+ \leqslant L_-$, we have

$$R_X(z) - R_X(z^*) = \{L_- - dU_+\} 1\{z=1\} + (1-d)U_+ 1\{z=-1\}.$$

and $R_{X,\phi_d}(z^*) = U_+ + L_-$. In this scenario, direct calculation yields

$$\begin{aligned} R_{-1}(\boldsymbol{X}) - R_{X,\phi_d}(z^*) &= \delta\left\{U_+/d - L_-\right\}, \\ R_1(\boldsymbol{X}) - R_{X,\phi_d}(z^*) &= \delta\left\{L_-/d - U_+\right\}, \\ R_0(\boldsymbol{X}) - R_{X,\phi_d}(z^*) &= 0. \end{aligned}$$

Thus, it can be verified that for $\delta > 0$, we have

$$R_X(z) - R_X(z^*) \leqslant \frac{d}{\delta}\left\{R_z(\boldsymbol{X}) - R_{X,\phi_d}(z^*)\right\};$$

for $\delta = 0$ and $d = 1$, we have

$$R_X(z) - R_X(z^*) \leqslant R_z(\boldsymbol{X}) - R_{X,\phi_d}(z^*).$$

When $dU_+ > L_-$, we have

$$R_X(z) - R_X(z^*) = \left\{U_+ - L_-\right\} 1\left\{z = -1\right\} + \left\{dU_+ - L_-\right\} 1\left\{z = 0\right\}.$$

and $R_{X,\phi_d}(z^*) = (1 + 1/d)L_-$. In this scenario, direct calculation yields

$$\begin{aligned} R_{-1}(\boldsymbol{X}) - R_{X,\phi_d}(z^*) &= U_+(1 + \delta/d) - L_-(\delta + 1/d), \\ R_1(\boldsymbol{X}) - R_{X,\phi_d}(z^*) &= 0, \\ R_0(\boldsymbol{X}) - R_{X,\phi_d}(z^*) &= (1 - \delta)\left\{U_+ - L_-/d\right\}. \end{aligned}$$

We will show that for $\delta \neq 0$, we have

$$R_X(z) - R_X(z^*) \leqslant \max\left\{\frac{d}{1-\delta}, \frac{d}{\delta(1+d)}\right\}\left\{R_z(\boldsymbol{X}) - R_{X,\phi_d}(z^*)\right\};$$

for $\delta = 0$ and $d = 1$, we have

$$R_X(z) - R_X(z^*) \leqslant R_z(\boldsymbol{X}) - R_{X,\phi_d}(z^*).$$

When $\delta = 0$ and $d = 1$, it is easy to verify that

$$R_X(z) - R_X(z^*) = R_z(\boldsymbol{X}) - R_{X,\phi_d}(z^*).$$

When $\delta \neq 0$, it suffices to verify that

$$\begin{aligned} U_+ - L_- &\leqslant \max\left\{\frac{d}{1-\delta}, \frac{d}{\delta(1+d)}\right\}\{U_+(1+\delta/d) - L_-(\delta + 1/d)\}, \\ dU_+ - L_- &\leqslant \max\left\{\frac{d}{1-\delta}, \frac{d}{\delta(1+d)}\right\}(1-\delta)\{U_+ - L_-/d\}. \end{aligned}$$

The second inequality is immediate from

$$dU_+ - L_- = \frac{d}{1-\delta}(1-\delta)\{U_+ - L_-/d\}.$$

To verify the first inequality, we calculate the following

$$\begin{aligned} & \frac{d}{\delta(1+d)}\{U_+(1+\delta/d) - L_-(\delta+1/d)\} - \{U_+ - L_-\} \\ = & \frac{1-\delta}{\delta(1+d)}\{dU_+ - L_-\}. \end{aligned}$$

Since $dU_+ > L_-$, we have that the first inequality holds for $\delta \neq 0$. Combining these results, we conclude the proof of Lemma 1.

In Part III of this proof, we will use Lemma 1 to prove Theorem 2. When $\delta \neq 0$, Lemma 1 shows that

When $LU \geqslant 0$, we have

$$R_X(z) - R_X(z^*) \leqslant \frac{1}{1-\delta} \left\{ R_z(\boldsymbol{X}) - R_{X,\phi_d}(z^*) \right\}.$$

When $LU < 0$, and $0 < \delta < 1$, we have

$$R_X(z) - R_X(z^*) \leqslant \max\left\{ \frac{d}{1-\delta}, \frac{d}{\delta(1+d)} \right\} \left\{ R_z(\boldsymbol{X}) - R_{X,\phi_d}(z^*) \right\};$$

. Thus, we have

$$R_X(z) - R_X(z^*) \leqslant \max\left\{ \frac{1}{1-\delta}, \frac{d}{\delta(1+d)} \right\} \left\{ R_z(\boldsymbol{X}) - R_{X,\phi_d}(z^*) \right\}.$$

Taking the expectation over X, we have

$$R(D) - R^* \leqslant \max\left\{ \frac{1}{1-\delta}, \frac{d}{\delta(1+d)} \right\} E\left[R_z(\boldsymbol{X}) - R_{X,\phi_d}(z^*) \right].$$

Notice that $R_z(\boldsymbol{X}) - R_{X,\phi_d}(z^*) \leqslant R_{X,\phi_d}(f) - R_{X,\phi_d}(z^*)$, thus we have

$$R(D) - R^* \leqslant \max\left\{ \frac{1}{1-\delta}, \frac{d}{\delta(1+d)} \right\} \left\{ R_{\phi_d}(f) - R^*_{\phi_d} \right\}.$$

When $\delta = 0$ and $d = 1$, Lemma 1 immediately shows that

$$R(D) - R^* \leqslant \left\{ R_{\phi_d}(f) - R^*_{\phi_d} \right\}.$$

□

3.2 Bound approximation errors: generalized geometric conditions

In this section, we introduce generalized geometric conditions that allows us to derive the approximation error, $a(\lambda_n)$, when we adopt the Gaussian RBF kernels. We assume that $L(\boldsymbol{x})$ and $U(\boldsymbol{x})$ are continuous functions of $\boldsymbol{x}$. In order to introduce the generalized geometric conditions, we define the classes of $\mathcal{X}$ by

$$\begin{aligned}
\mathcal{X}_+ &:= \{\boldsymbol{x} \in \mathcal{X} : L_-(\boldsymbol{X}) < dU_+(\boldsymbol{X})\}, \\
\mathcal{X}_- &:= \{\boldsymbol{x} \in \mathcal{X} : U_+(\boldsymbol{X}) < dL_-(\boldsymbol{X})\}, \\
\mathcal{X}_0 &:= \mathcal{X}_+^c \cap \mathcal{X}_-^c \cap \mathcal{X}.
\end{aligned}$$

Based on these defined classes, we define three distance functions

$$\tau_{\boldsymbol{x},+} := \begin{cases} d(\boldsymbol{x}, \mathcal{X}_- \cup \mathcal{X}_0), & \text{if } \boldsymbol{x} \in \mathcal{X}_+ \\ d(\boldsymbol{x}, \mathcal{X}_+), & \text{otherwise,} \end{cases}$$

$$\tau_{\boldsymbol{x},-} := \begin{cases} d(\boldsymbol{x}, \mathcal{X}_+ \cup \mathcal{X}_0), & \text{if } \boldsymbol{x} \in \mathcal{X}_- \\ d(\boldsymbol{x}, \mathcal{X}_-), & \text{otherwise,} \end{cases}$$

and

$$\tau_{\boldsymbol{x},0} := \begin{cases} \tau_{\boldsymbol{x},+}, & \text{if } \boldsymbol{x} \in \mathcal{X}_- \\ \tau_{\boldsymbol{x},-}, & \text{if } \boldsymbol{x} \in \mathcal{X}_+ \\ \max\{\tau_{\boldsymbol{x},-}, \tau_{\boldsymbol{x},+}\}, & \text{otherwise,} \end{cases}$$

where $d(\boldsymbol{x}, A)$ represents the Euclidean norm of $\boldsymbol{x}$ to the closure of $A \subset \mathcal{X}$. The distance functions $\tau_{\boldsymbol{x},+}$ and $\tau_{\boldsymbol{x},-}$ give the distance between a data point $\boldsymbol{x}$ and two decision boundaries, i.e., $\{L_-(\boldsymbol{X}) = dU_+(\boldsymbol{X})\}$ and $\{U_+(\boldsymbol{X}) = dL_-(\boldsymbol{X})\}$. The distance function $\tau_{\boldsymbol{x},0}$ is defined as the maximum distance between a data point $\boldsymbol{x}$ and two decision boundaries. By the definitions of $\tau_{\boldsymbol{x},0}$, we have that $\tau_{\boldsymbol{x},0} \geqslant \tau_{\boldsymbol{x},+}$, and $\tau_{\boldsymbol{x},0} \geqslant \tau_{\boldsymbol{x},-}$. Additionally, $\tau_{\boldsymbol{x},0} = 0$ implies that $\tau_{\boldsymbol{x},+} = \tau_{\boldsymbol{x},-} = 0$, and thus $L_-(\boldsymbol{X}) = U_+(\boldsymbol{X}) = 0$.

Now we present the generalized geometric conditions.

Definition 1. *Let $\mathcal{X} \subset \mathbb{R}^p$ be compact and P_X be a probability measure on $\mathcal{X}$. We say that P_X has generalized geometric noise exponent $\alpha > 0$ under the deferred option parameter d if there exists a constant $C_\alpha > 0$ such that for $t > 0$,*

$$\begin{aligned}
\int_{\mathcal{X}_+ \cup \mathcal{X}_0} |U_+(\boldsymbol{X}) - L_-(\boldsymbol{X})/d| \exp\left(-\frac{\tau^2_{\boldsymbol{x},+}}{t}\right) P_X(\mathrm{d}\boldsymbol{x}) &\leqslant C_\alpha t^{\alpha p/2}, \\
\int_{\mathcal{X}_- \cup \mathcal{X}_0} |L_-(\boldsymbol{X}) - U_+(\boldsymbol{X})/d| \exp\left(-\frac{\tau^2_{\boldsymbol{x},-}}{t}\right) P_X(\mathrm{d}\boldsymbol{x}) &\leqslant C_\alpha t^{\alpha p/2}, \\
\int_{\mathcal{X}_+ \cup \mathcal{X}_-} \{L_-(\boldsymbol{X}) + U_+(\boldsymbol{X})\} \exp\left(-\frac{\tau^2_{\boldsymbol{x},0}}{t}\right) P_X(\mathrm{d}\boldsymbol{x}) &\leqslant C_\alpha t^{\alpha p/2}. \qquad (2)
\end{aligned}$$

We say that P_X has generalized geometric noise exponent ∞ under the deferred option parameter d if it has generalized geometric noise exponent α for all $\alpha > 0$.

Remark 1. *The geometric condition associated with geometric noise exponent α and constant C_α in a binary classification problem is*

$$\int_{\mathcal{X}} |2\eta(\boldsymbol{X}) - 1| \exp\left(-\frac{\tau^2_{\boldsymbol{x}}}{t}\right) P_X(\mathrm{d}\boldsymbol{x}) \leqslant C_\alpha t^{\alpha p/2}, \qquad (3)$$

where $\eta(\boldsymbol{X}) = P(Y = 1 \mid \boldsymbol{X})$ and Y is a binary label to predict in this binary classification problem.

In classification problem,

$$\tau_{\boldsymbol{x}} := \begin{cases} d(\boldsymbol{x}, \mathcal{X}_+ \cup \mathcal{X}_0), & \text{if } \boldsymbol{x} \in \mathcal{X}_- \\ d(\boldsymbol{x}, \mathcal{X}_- \cup \mathcal{X}_0), & \text{if } \boldsymbol{x} \in \mathcal{X}_+ \\ 0, & \text{otherwise,} \end{cases}$$

where $\mathcal{X}_- = \{\boldsymbol{x} \in \mathcal{X} : \eta(\boldsymbol{x}) < 1/2\}$, $\mathcal{X}_+ = \{\boldsymbol{x} \in \mathcal{X} : \eta(\boldsymbol{x}) > 1/2\}$, *and* $\mathcal{X}_0 = \{x \in \mathcal{X} : \eta(\boldsymbol{x}) = 1/2\}$. *Steinwart et al. (2007) introduced this geometric condition to derive an upper bound for approximation error when using Gaussian RBF kernels and a hinge loss function. The integral condition 3 describes the concentration of the measure* $|2\eta(\boldsymbol{X}) - 1| \, \mathrm{d}P_X$ *near the decision boundary. Our conditions can be considered as a generalization of this geometric condition to accommodate the deferred option.*

The generalized geometric conditions do not require any kind of smoothness assumptions. Instead, the first two integral conditions (2) describe the concentrations of the measure $|U_+(\boldsymbol{X}) - L_-(\boldsymbol{X})/d| \, \mathrm{d}P_X$ near the decision boundary $\{U_+(\boldsymbol{X}) = L_-(\boldsymbol{X})/d\}$ and the measure $|L_-(\boldsymbol{X}) - U_+(\boldsymbol{X})/d| \, \mathrm{d}P_X$ near the decision boundary $\{L_-(\boldsymbol{X}) = U_+(\boldsymbol{X})/d\}$. The third integral condition describes the concentration of the measure $L_-(\boldsymbol{X}) + U_+(\boldsymbol{X})$ near the identified subgroup of no treatment effects, i.e., $\{L(\boldsymbol{X}) = U(\boldsymbol{X}) = 0\}$.

Theorem 3. *Let* $\sigma > 0$, $\mathcal{X}$ *be the closed unit ball. Furthermore,* P_X *has generalized geometric noise exponent* $\alpha > 0$ *with a constant* C_α *in* (2). *Then there is a constant* $c_d > 0$ *depending only on the dimension* p *such that for all* $\lambda > 0$, *we have*

$$a_\sigma(\lambda) \leqslant c_d(\sigma^p \lambda + C_\alpha (2d)^{\alpha p/2} \sigma^{-\alpha p}).$$

Proof. Let $H_\sigma(\mathbb{R}^p)$ be the RKHS of the Gaussian RBF kernel with parameter σ. Then, by (Steinwart

et al. 2006), the linear operator $V_\sigma : L_2(\mathbb{R}^p) \to H_\sigma(\mathbb{R}^p)$ defined by

$$V_\sigma g(\boldsymbol{X}) = \frac{(2\sigma)^{d/2}}{\pi^{d/4}} \int_{\mathbb{R}^p} \exp\left\{-2\sigma^2 \left\|x - y\right\|\right\} g(y)\mathrm{d}y, g \in L_2(\mathbb{R}^p), x \in \mathbb{R}^p,$$

is an isometric isomorphism. Consequently, we have

$$a(\lambda) = \inf_{g \in L_2(\mathbb{R}^p)} \left(\lambda \left\|g\right\|^2_{L_2(\mathbb{R}^p)} + R_{\phi_d}(V_\sigma g) - R^*_{\phi_d}\right), \lambda > 0.$$

In the following, we will estimate the right-hand side by a judicious choice of g. To start with, we need the following lemma, which ensures that all balls of the form $B(x, \tau_x)$ are contained in the (enlarged) support.

Lemma 2. *Let $\mathcal{X}$ be a closed unit ball of $\mathbb{R}^p$ and define $\eta(\boldsymbol{X}) = L_-(\boldsymbol{X})/U_+(\boldsymbol{X})$. Then we have*

$$\begin{aligned}
\mathcal{X}_+ &= \{x \in \mathcal{X} : \eta(\boldsymbol{X}) < d\}, \\
\mathcal{X}_- &= \{x \in \mathcal{X} : \eta(\boldsymbol{X}) > 1/d\}, \\
\mathcal{X}_0 &= \mathcal{X}_+^c \cap \mathcal{X}_-^c \cap \mathcal{X}.
\end{aligned}$$

On $\widetilde{\mathcal{X}} := 3\mathcal{X}$, we define

$$\widetilde{\eta}(\boldsymbol{X}) := \begin{cases} \eta(\boldsymbol{X}), & \text{if } \left\|x\right\|_2 \leqslant 1 \\ \eta(x/\left\|x\right\|_2), & \text{otherwise.} \end{cases}$$

We also write

$$\begin{aligned}\widetilde{\mathcal{X}}_+ &= \{x \in \mathcal{X} : \widetilde{\eta}(\boldsymbol{X}) < d\}, \\ \widetilde{\mathcal{X}}_- &= \{x \in \mathcal{X} : \widetilde{\eta}(\boldsymbol{X}) > 1/d\}, \\ \widetilde{\mathcal{X}}_0 &= \widetilde{\mathcal{X}}_+^c \cap \widetilde{\mathcal{X}}_-^c \cap \widetilde{\mathcal{X}}.\end{aligned}$$

Then for $x \in \mathcal{X}_+$, we have $B(x, \tau_{\boldsymbol{x},+}) \subset \widetilde{\mathcal{X}}_+$ and $B(x, \tau_{\boldsymbol{x},0}) \subset \widetilde{\mathcal{X}}_+ \cup \widetilde{\mathcal{X}}_0$; for $x \in \mathcal{X}_-$, we have $B(x, \tau_{\boldsymbol{x},-}) \subset \widetilde{\mathcal{X}}_-$ and $B(x, \tau_{\boldsymbol{x},0}) \subset \widetilde{\mathcal{X}}_- \cup \widetilde{\mathcal{X}}_0$.

Proof. Proof follows the proof of Lemma 4.1 in Steinwart et al. (2007). □

We fix a measurable function $\widetilde{f}$ that satisfies $\widetilde{f} = 1$ on $\widetilde{\mathcal{X}}_+$; $\widetilde{f} = -1$ on $\widetilde{\mathcal{X}}_-$; $\widetilde{f} = 0$, otherwise. For $g := (\sigma^2/\pi)^{d/4}\widetilde{f}$, we then immediately obtain

$$\|g\|_{L_2(\mathbb{R}^p)} \leqslant (81\sigma^2/\pi)^{p/4}\theta(p),$$

where $\theta(p)$ denotes the volume of $\mathcal{X}$. Moreover, we have that $-1 \leqslant V_\sigma g \leqslant 1$.

Next, we will estimate $R_{\phi_d}(f) - R^*_{\phi_d}$ by a arbitrary choice of f which takes values in $[-1, 1]$. By direct computation, we have

$$\begin{aligned}R_{\phi_d}(f) - R^*_{\phi_d} \leqslant\ & E\left[|L_-(\boldsymbol{X})/d - U_+(\boldsymbol{X})|\,|f - f_d^*|1\left\{f_d^* \neq -1, 0 < f \leqslant 1\right\}\right] \\ & +(1 - 1/d)E\left[|L_-(\boldsymbol{X}) + U_+(\boldsymbol{X})|\,|f|1\left\{f_d^* = 1, -1 < f \leqslant 0\right\}\right] \\ & +E\left[|L_-(\boldsymbol{X}) - U_+(\boldsymbol{X})/d|\,|f - f_d^*|1\left\{f_d^* \neq 1, -1 < f \leqslant 0\right\}\right] \\ & +(1 - 1/d)E\left[|L_-(\boldsymbol{X}) + U_+(\boldsymbol{X})|\,|f|1\left\{f_d^* = -1, 0 < f \leqslant 1\right\}\right].\end{aligned}$$

Based on this inequality, we first bound

$$E\left[|L_-(\boldsymbol{X})/d - U_+(\boldsymbol{X})|\,|f - f_d^*|1\left\{f_d^* \neq -1, 0 < f \leqslant 1\right\}\right].$$

Following the proof of Theorem 2.7 in Steinwart et al. (2007), we conclude that for $\mathcal{X}_+$, we have

$$|V_\sigma g(\boldsymbol{X}) - 1| \leqslant 8\exp\{-\sigma^2\tau_{\boldsymbol{x},+}^2/2d\}.$$

For $x \in \mathcal{X}_0$, we also have

$$\begin{aligned}
V_\sigma g(\boldsymbol{X}) &= \left(\frac{2\sigma^2}{\pi}\right)^{d/2}\int_{\mathbb{R}^p}\exp\left\{-2\sigma^2\left\|x-y\right\|_2^2\right\}\widetilde{f}(y)\mathrm{d}y \\
&= \left(\frac{2\sigma^2}{\pi}\right)^{d/2}\int_{\mathbb{R}^p}\exp\left\{-2\sigma^2\left\|x-y\right\|_2^2\right\}(\widetilde{f}(y)-1)\mathrm{d}y + 1 \\
&\leqslant \left(\frac{2\sigma^2}{\pi}\right)^{d/2}\int_{B(x,\tau_{\boldsymbol{x},+})}\exp\left\{-2\sigma^2\left\|x-y\right\|_2^2\right\}(\widetilde{f}(y)-1)\mathrm{d}y + 1 \\
&\leqslant -\left(\frac{2\sigma^2}{\pi}\right)^{d/2}\int_{B(x,\tau_{\boldsymbol{x},+})}\exp\left\{-2\sigma^2\left\|x-y\right\|_2^2\right\}\mathrm{d}y + 1 \\
&= P_{\gamma_\sigma}(|u| \geqslant \tau_{\boldsymbol{x},+}) \\
&\leqslant 4\exp\{-\sigma^2\tau_{\boldsymbol{x},+}^2/2d\}.
\end{aligned}$$

Thus, we conclude that

$$|V_\sigma g(\boldsymbol{X})| \leqslant 4\exp\{-\sigma^2\tau_{\boldsymbol{x},+}^2/2d\}$$

if $V_\sigma g(\boldsymbol{X}) > 0$. Consequently, by the generalized geometric noise assumption, we have

$$\begin{aligned}
& E\left[|L_-(\boldsymbol{X})/d - U_+(\boldsymbol{X})|\,|f - f_d^*|1\left\{f_d^* \neq -1, 0 < f \leqslant 1\right\}\right] \\
\leqslant\ & E\left[|L_-(\boldsymbol{X})/d - U_+(\boldsymbol{X})|\, 8\exp\{-\sigma^2\tau_{\boldsymbol{x},+}^2/2d\}1\left\{f_d^* \neq -1, 0 < f \leqslant 1\right\}\right] \\
=\ & 8\int_{\mathcal{X}_+\cup\mathcal{X}_0} |L_-(\boldsymbol{X})/d - U_+(\boldsymbol{X})|\exp\{-\sigma^2\tau_{\boldsymbol{x},+}^2/2d\}P_X(\mathrm{d}x) \\
\leqslant\ & 8C_\alpha(2d/\sigma^2)^{\alpha p/2}.
\end{aligned}$$

Similarly, we can show that

$$E\left[|L_-(\boldsymbol{X}) - U_+(\boldsymbol{X})/d|\,|f - f_d^*|1\left\{f_d^* \neq 1, -1 < f \leqslant 0\right\}\right] \leqslant 8C_\alpha(2d/\sigma^2)^{\alpha p/2}.$$

Next, we consider a bound for

$$\begin{aligned}
& (1-1/d)E\left[|L_-(\boldsymbol{X}) + U_+(\boldsymbol{X})|\,|f|1\left\{f_d^* = 1, -1 < f \leqslant 0\right\}\right] \\
& +(1-1/d)E\left[|L_-(\boldsymbol{X}) + U_+(\boldsymbol{X})|\,|f|1\left\{f_d^* = -1, 0 < f \leqslant 1\right\}\right].
\end{aligned}$$

When $x \in \mathcal{X}_+$, we have

$$\begin{aligned}
V_\sigma g(\boldsymbol{X}) &= \left(\frac{2\sigma^2}{\pi}\right)^{d/2} \int_{\mathbb{R}^p} \exp\left\{-2\sigma^2 \|x-y\|_2^2\right\} \widetilde{f}(y)\mathrm{d}y \\
&= \left(\frac{2\sigma^2}{\pi}\right)^{d/2} \int_{\mathbb{R}^p} \exp\left\{-2\sigma^2 \|x-y\|_2^2\right\} (\widetilde{f}(y)+1)\mathrm{d}y - 1 \\
&\geqslant \left(\frac{2\sigma^2}{\pi}\right)^{d/2} \int_{B(x,\tau_{\boldsymbol{x},+})} \exp\left\{-2\sigma^2 \|x-y\|_2^2\right\} (\widetilde{f}(y)+1)\mathrm{d}y \\
&\quad + \left(\frac{2\sigma^2}{\pi}\right)^{d/2} \int_{B(x,\tau_{\boldsymbol{x},0})/B(x,\tau_{\boldsymbol{x},+})} \exp\left\{-2\sigma^2 \|x-y\|_2^2\right\} (\widetilde{f}(y)+1) - 1 \\
&\geqslant 2\left(\frac{2\sigma^2}{\pi}\right)^{d/2} \int_{B(x,\tau_{\boldsymbol{x},+})} \exp\left\{-2\sigma^2 \|x-y\|_2^2\right\} \mathrm{d}y \\
&\quad + \left(\frac{2\sigma^2}{\pi}\right)^{d/2} \int_{B(x,\tau_{\boldsymbol{x},0})/B(x,\tau_{\boldsymbol{x},+})} \exp\left\{-2\sigma^2 \|x-y\|_2^2\right\} - 1 \\
&= P_{\gamma_\sigma}(|u| \leqslant \tau_{\boldsymbol{x},+}) - P_{\gamma_\sigma}(|u| \geqslant \tau_{\boldsymbol{x},0}).
\end{aligned}$$

Thus, for any x such that $-1 < V_\sigma g(\boldsymbol{X}) \leqslant 0$, we have

$$V_\sigma g(\boldsymbol{X}) \geqslant -4\exp\{-\sigma^2\tau_{\boldsymbol{x},0}^2/2d\}.$$

Therefore, we have

$$\begin{aligned}
&(1-1/d)E\left[|L_-(\boldsymbol{X})+U_+(\boldsymbol{X})|\,|f|1\left\{f_d^*=1, -1<f\leqslant 0\right\}\right] \\
\leqslant\ &(1-1/d)E\left[|L_-(\boldsymbol{X})+U_+(\boldsymbol{X})|\,4\exp\{-\sigma^2\tau_{\boldsymbol{x},0}^2/2d\}1\left\{f_d^*=1, -1<f\leqslant 0\right\}\right].
\end{aligned}$$

Similar results hold when $x \in \mathcal{X}_-$. Hence, we have

$$\begin{aligned}
& (1-1/d)E\left[|L_-(\boldsymbol{X})+U_+(\boldsymbol{X})|\,|f|1\left\{f_d^*=1,-1<f\leqslant 0\right\}\right] \\
& +(1-1/d)E\left[|L_-(\boldsymbol{X})+U_+(\boldsymbol{X})|\,|f|1\left\{f_d^*=-1,0<f\leqslant 1\right\}\right] \\
\leqslant\ & (1-1/d)E\left[|L_-(\boldsymbol{X})+U_+(\boldsymbol{X})|\,4\exp\{-\sigma^2\tau_{\boldsymbol{x},0}^2/2d\}\right] \\
\leqslant\ & 4C_\alpha(2d/\sigma^2)^{\alpha p/2}.
\end{aligned}$$

Combining the two bounds, we have

$$R_{\phi_d}(f)-R_{\phi_d}^*\leqslant 20C_\alpha(2d/\sigma^2)^{\alpha p/2}.$$

□

A simple regularity condition near decision boundaries can guarantee the generalized geometric condition.

Definition 2. *We say that P_X has a generalized envelope of order γ if there exists a constant c_γ such that for almost all $x \in \mathcal{X}$, we have*

$$\begin{aligned}
|U_+(\boldsymbol{X})-L_-(\boldsymbol{X})/d| &\leqslant c_\gamma\tau_{\boldsymbol{x},+}^\gamma, \\
|L_-(\boldsymbol{X})-U_+(\boldsymbol{X})/d| &\leqslant c_\gamma\tau_{\boldsymbol{x},-}^\gamma \\
|U_+(\boldsymbol{X})+L_-(\boldsymbol{X})| &\leqslant c_\gamma\tau_{\boldsymbol{x},0}^\gamma.
\end{aligned}$$

The generalized envelope forms a multiple of the envelope defined by $\tau_{\boldsymbol{x},+}^\gamma$ and $-\tau_{\boldsymbol{x},+}^\gamma$, $\tau_{\boldsymbol{x},-}^\gamma$ and $-\tau_{\boldsymbol{x},-}^\gamma$, $\tau_{\boldsymbol{x},0}^\gamma$ and $-\tau_{\boldsymbol{x},0}^\gamma$ for the graph of $(\boldsymbol{x},|U_+(\boldsymbol{X})-L_-(\boldsymbol{X})/d|)$, $(\boldsymbol{x},|L_-(\boldsymbol{X})-U_+(\boldsymbol{X})/d|)$, and $(\boldsymbol{x},|U_+(\boldsymbol{X})+L_-(\boldsymbol{X})|)$. Under this condition, the $L(\boldsymbol{X})$ and $U(\boldsymbol{X})$ can behave irregularly away from the decision boundary but cannot be discontinues when crossing these boundaries. We also define the

Tsybakov noise exponent in the following.

Definition 3. *Let $0 \leqslant q \leqslant +\infty$ and P_X be a probability measure on $\mathcal{X}$. We say that P_X has Tsybakov noise exponent q if there exists a constant $C_q > 0$ such that for all sufficiently small $t > 0$ we have*

$$\begin{aligned} P\left\{0 \neq |U_+(\boldsymbol{X}) - L_-(\boldsymbol{X})/d| \leqslant t\right\} &\leqslant C_q t^q, \\ P\left\{0 \neq |L_-(\boldsymbol{X}) - U_+(\boldsymbol{X})/d| \leqslant t\right\} &\leqslant C_q t^q, \\ P\left\{0 \neq |U_+(\boldsymbol{X}) + L_-(\boldsymbol{X})| \leqslant t\right\} &\leqslant C_q t^q. \end{aligned}$$

Theorem 4. *Let $\mathcal{X}$ be compact in $\mathbb{R}^p$ and P_X be a distribution that has an envelope of order $\gamma > 0$ and a Tsybakov noise exponent $q \in [0, +\infty)$. Then P_X has geometric noise exponent $(q+1)\gamma p^{-1}$ if $q \geqslant 1$, and geometric noise exponent α for all $\alpha < (q+1)\gamma p^{-1}$ otherwise.*

Proof. We will follow the logic of Theorem 2.6 in Steinwart et al. (2007), where the Hölder type inequalities in Lorentz spaces are adopted. We take the inequality

$$\int_{\mathcal{X}_+ \cup \mathcal{X}_0} |U_+(\boldsymbol{X}) - L_-(\boldsymbol{X})/d| \exp\left(-\frac{\tau^2_{\boldsymbol{x},+}}{t}\right) P_X(\mathrm{d}\boldsymbol{x}) \leqslant C_\alpha t^{\alpha p/2}.$$

as an example. First, we consider the case where $q \geqslant 1$. When $q \geqslant 1$, we use the Hölder inequality for Lorentz spaces (O'Neil 1963), i.e.,

$$\|fg\|_1 \leqslant \|f\|_{q,\infty} \|g\|_{q',1},$$

for all $f \in L_{q,\infty}$, $g \in L_{q',1}$ and q' defined by $1/q + 1/q' = 1$. This inequity yields

$$\begin{aligned} & E\left[|U_+(\boldsymbol{X}) - L_-(\boldsymbol{X})/d| \exp\left(-\tau^2_{\boldsymbol{x},+}/t\right)\right] \\ \leqslant\ & \| \, |U_+(\boldsymbol{x}) - L_-(\boldsymbol{x})/d|^{-1} \, \|_{q,\infty} \| \left\{U_+(\boldsymbol{x}) - L_-(\boldsymbol{x})/d\right\}^2 \exp\left(-\tau^2_{\boldsymbol{x},+}/t\right) \|_{q',1} \\ \leqslant\ & C \| \left\{U_+(\boldsymbol{x}) - L_-(\boldsymbol{x})/d\right\}^2 \exp\left(-\left(|U_+(\boldsymbol{x}) - L_-(\boldsymbol{x})/d| / C_q\right)^{2/\gamma} / t\right) \|_{q',1}, \end{aligned}$$

where the last inequality uses the Tsybakov assumption (defined in Definition 3) and that P_X has an envelope of order $\gamma > 0$. Let $h(\boldsymbol{x}) := \{|dU_+ - L_-|/(1+d)\}^{-1}$, and $b := t(C_q)^{2/\gamma}$ so that

$$\{U_+(\boldsymbol{x}) - L_-(\boldsymbol{x})/d\}^2 \exp\left(-\left(|U_+(\boldsymbol{x}) - L_-(\boldsymbol{x})/d|/C_q\right)^{2/\gamma}/t\right) = g_d(h(\boldsymbol{x})),$$

where $g_d(s) := \left(\frac{ds}{1+d}\right)^{-2} \exp\left\{-\left(\frac{ds}{1+d}\right)^{-2/\gamma}/b\right\}$ for all $s \geqslant (1+d)/d$. It can be verified that $g(s)$ in strictly increasing on $[(1+d)/d, \infty]$ if $0 < b \leqslant \frac{2}{3\gamma}$, and hence we can extend g to a strictly increasing, continuous and invertible function on $[0, \infty]$ in this case. We also denote such an extension as g_d. Then for this extension we have

$$P(g_d \circ h > \tau) = P(h > g_d^{-1}(\tau)).$$

Now for a function f, we denote its nonincreasing rearrangement as f^*, i.e., $f^*(u) := \inf\{\sigma \geqslant 0 : P(f > \sigma) \leqslant u\}$ for $u > 0$. The nonincreasing rearrangement f^* can be used to define Lorentz norms (see Steinwart et al. (2007); Bennett and Sharpley (1988)). Inequality 3 implies that $P\left(h \geqslant \frac{1+d}{d}\left\{\frac{C_q}{u}\right\}^{1/q}\right) \leqslant u$ for all $u > 0$. Therefore, we find

$$h^*(u) \leqslant \inf\{\sigma \geqslant 0 : P(h > \sigma) \leqslant u\} \leqslant \frac{1+d}{d}\left\{\frac{C_q}{u}\right\}^{1/q}.$$

Since $(g_d \circ h)^* = g_d \circ h^*$ and g_d is increasing, we have

$$(g_d \circ h)^*(u) \leqslant g_d\left(\frac{1+d}{d}\left\{\frac{C_q}{u}\right\}^{1/q}\right)$$

for all $0 < u < 1$. Following the proof of Theorem 2.6 in Steinwart et al. (2007), we have

$$(g_d \circ h)^*(u) \lesssim b^{\hat{\alpha}} \frac{u^{2/q(1-\hat{\alpha}/\gamma)}}{\log^2((u/C_q)^{2/q\gamma} b^{-1}) + 1}$$

for $u > 0$ and any fixed $\hat{\alpha} > 0$ (without any loss of generality, we assume that $C_q \geqslant 1$). Then, we choose $\hat{\alpha} = \gamma\frac{q+1}{2}$ and $q' = q/(1-q)$ such that

$$\|g_d \circ h\| = \int_0^\infty u^{1/q'-1}(g_d \circ h)^*(u)du \lesssim t^{\gamma(q+1)/2},$$

since $(g_d \circ h)^*(u) = 0$ for $u \geqslant 1$. Thus, the assertion holds for $q \geqslant 1$. Let's consider the case $0 \leqslant q < 1$ where the Hölder inequality in Lorentz spaces cannot be used. In this case, for all $t, \tau \geqslant 0$, we have

$$\begin{aligned}
& E\left[|U_+(\boldsymbol{X}) - L_-(\boldsymbol{X})/d| \exp\left(-\tau^2_{\boldsymbol{x},+}/t\right)\right] \\
= & \int_{|U_+(\boldsymbol{X})-L_-(\boldsymbol{X})/d|\leqslant\tau} |U_+(\boldsymbol{X}) - L_-(\boldsymbol{X})/d| \exp\left(-\tau^2_{\boldsymbol{x},+}/t\right) P_x(dx) \\
& + \int_{|U_+(\boldsymbol{X})-L_-(\boldsymbol{X})/d|>\tau} |U_+(\boldsymbol{X}) - L_-(\boldsymbol{X})/d| \exp\left(-\tau^2_{\boldsymbol{x},+}/t\right) P_x(dx) \\
\leqslant & C_q\tau^{q+1} + d^{-1}\exp\left\{-\left(\tau/c_\gamma\right)^{2/\gamma} t^{-1}\right\},
\end{aligned}$$

where we used the Tsybakov assumption3 and the fact that P_x has an envelope of order γ. We can choose τ such that $\tau^{q+1} = \exp\left\{-\left(\tau/c_\gamma\right)^{2/\gamma} t^{-1}\right\}$. This choice implies that for t sufficiently small,

$$\tau \leqslant \left(\frac{\hat{\alpha}\gamma}{2}\right)^{\gamma/2}\left(t\log\frac{1}{\hat{\alpha}t}\right)^{\gamma/2},$$

where $\hat{\alpha} = c_\gamma^{2/\gamma}(q+1)$. This yields the assertion for the case $0 \leqslant q < 1$. □

In the following, we provide a further condition for the Tsybakov condition in (3).

Theorem 5. *If $|U_+(\boldsymbol{X}) - L_-(\boldsymbol{X})/d|$, $|L_-(\boldsymbol{X}) - U_+(\boldsymbol{X})/d|$, and $|U_+(\boldsymbol{X}) + L_-(\boldsymbol{X})|$ have bounded density function on $(0,\infty)$, then Tsybakov condition in (3) holds with $q = 1$.*

Proof. We take $|U_+(\boldsymbol{X}) - L_-(\boldsymbol{X})/d|$ as an example. Let $f_{|U_+(\boldsymbol{x})-L_-(\boldsymbol{x})/d|}(u)$ be the pd.f. of $|U_+(\boldsymbol{X}) -$

$L_-(\boldsymbol{X})/d|$. We can directly calculate that

$$P\left\{0 \neq |U_+(\boldsymbol{X}) - L_-(\boldsymbol{X})/d| \leqslant t\right\} \leqslant \int_{0<u\leqslant t} f_{|U_+(\boldsymbol{x})-L_-(\boldsymbol{x})/d|}(u)du \leqslant Mt,$$

where M is the upper bound of $f_{|U_+(\boldsymbol{x})-L_-(\boldsymbol{x})/d|}(u)$ on $(0,\infty)$. □

In the following, we also provide a sufficient condition for the generalized envelope condition.

Theorem 6. *If $U_+(\boldsymbol{X}) - L_-(\boldsymbol{X})/d$, $L_-(\boldsymbol{X}) - U_+(\boldsymbol{X})/d$, and $U_+(\boldsymbol{X}) + L_-(\boldsymbol{X})$ are Hölder continuous with an order of γ, then the generalized envelope condition in (2) holds with γ.*

Proof. We will take $|U_+(\boldsymbol{X}) - L_-(\boldsymbol{X})/d|$ as an example; others follow similarly. For any $\boldsymbol{x} \in \mathcal{X}_+$ and any $\boldsymbol{x}' \in \mathcal{X}_- \cup \mathcal{X}_0$, by Hölder continuous, we have

$$|\left\{U_+(\boldsymbol{x}) - L_-(\boldsymbol{x})/d\right\} - \left\{U_+(\boldsymbol{x}') - L_-(\boldsymbol{x}')/d\right\}| \leqslant Md^\gamma(\boldsymbol{x},\boldsymbol{x}'),$$

for some constant M. Since $\boldsymbol{x} \in \mathcal{X}_+$ and $\boldsymbol{x}' \in \mathcal{X}_- \cup \mathcal{X}_0$, we have

$$|U_+(\boldsymbol{x}) - L_-(\boldsymbol{x})/d| \leqslant |\left\{U_+(\boldsymbol{x}) - L_-(\boldsymbol{x})/d\right\} - \left\{U_+(\boldsymbol{x}') - L_-(\boldsymbol{x}')/d\right\}| \leqslant Md^\gamma(\boldsymbol{x},\boldsymbol{x}').$$

Thus, we have

$$|U_+(\boldsymbol{x}) - L_-(\boldsymbol{x})/d| \leqslant Md^\gamma(\boldsymbol{x}, \mathcal{X}_- \cup \mathcal{X}_0) = M\tau^\gamma_{\boldsymbol{x},+}.$$

The proof above implies that

$$|U_+(\boldsymbol{x}') - L_-(\boldsymbol{x}')/d| \leqslant Md^\gamma(\boldsymbol{x}', \mathcal{X}_+) = M\tau^\gamma_{\boldsymbol{x}',+}.$$

This concludes the proof. □

From the theorems above, we can see that the Hölder continuity and bounded density functions are sufficient conditions for the proposed generalized geometric conditions. Thus, the generalized geometric conditions are weaker than the smoothness assumptions on the distribution or density functions.

3.3 Generalization Error

Assume

C1) There exists $0 < \epsilon < 1$ such that $\epsilon \leqslant \pi_z(\boldsymbol{X}) \leqslant 1 - \epsilon$ with probability 1.

C2) Suppose that for some $\beta_\pi, \beta_p > 0$, $\sup_{\boldsymbol{X}} |\widehat{\pi}_z(\boldsymbol{X}) - \pi_z(\boldsymbol{X})| = O_p(n^{-\beta_\pi})$ and $\sup_{\boldsymbol{X}} \left|\widehat{p}_{y,a|z}(\boldsymbol{X}) - p_{y,a|z}(\boldsymbol{X})\right| = O_p(n^{-\beta_p})$ for all y, a, and z.

Theorem 7. *Assume the assumptions C1) and C2). Let P_X has generalized geometric noise exponent α and boundary concentration exponent γ, then for any $\delta > 0$, $0 < v < 2$, we have*

$$\begin{aligned} & R\left\{D(X; \widehat{f}_d, \delta)\right\} - R^* \\ = \; & O_p\Big(\sigma_n^{(2-v)(1+\delta)p/(2+v)} \lambda^{-v/(v+2)} n^{-2/(2+v)} + n^{-1/2} \sigma_n^{(1-v/2)(1+\delta)p/2} \lambda_n^{-1/2} + n^{-(1+\gamma)\beta_p} \lambda_n^{-1/2} \\ & + n^{-\beta_\pi - \beta_p} \lambda_n^{-1/2} + c_d(\sigma_n^p \lambda_n + C_\alpha (2d)^{\alpha p/2} \sigma_n^{-\alpha p})\Big). \end{aligned}$$

Corollary 1. *Assume the assumptions C1) and C2). Let P_X has generalized geometric noise exponent α and boundary concentration exponent γ, then for any $\delta > 0$, $0 < v < 2$, with $\sigma_n = \lambda^{-1/(1+\alpha)p}$, we have*

$$\begin{aligned} & R\left(D(X; \widehat{f}_d, \delta)\right) - R^* \\ = \; & O_p\left(\lambda_n^{-\frac{v}{v+2} - \frac{(2-v)(1+\delta)}{(2+v)(1+\alpha)}} n^{-2/(2+v)} + \lambda_n^{-\frac{1}{2} - \frac{(2-v)(1+\delta)}{4(1+\alpha)}} n^{-1/2} + n^{-(1+\gamma)\beta_p} \lambda_n^{-1/2} + n^{-(\beta_\pi + \beta_p)} \lambda_n^{-1/2} + \lambda_n^{\frac{\alpha}{1+\alpha}}\right). \end{aligned}$$

Proof. By Lemma 3, with a large probability, there exists a constant M such that

$$\left\|\widehat{f}_d\right\|_{\mathcal{H}} \leqslant M\lambda_n^{-1/2},$$

and

$$\left\|f^*_{d,\lambda_n}\right\|_{\mathcal{H}} \leqslant M\lambda_n^{-1/2},$$

where f^*_{d,λ_n} is the minimizer of $\lambda_n \|f\|^2_{\mathcal{H}} + R_{\phi_d}(f)$.

To simplify our proof, we will choose $\delta = d/(1+d)$. By Theorem 2, we have

$$R\left\{D(X;\widehat{f}_d,\delta)\right\} - R^* \leqslant (1+d)\left\{R_{\phi_d}(\widehat{f}_d) - R^*_{\phi_d}\right\}.$$

Notice that

$$\begin{aligned} R_{\phi_d}(\widehat{f}_d) - R^*_{\phi_d} &\leqslant R_{\phi_d}(\widehat{f}_d) + \lambda_n \left\|\widehat{f}_d\right\|^2_{\mathcal{H}} - R^*_{\phi_d} \\ &= R_{\phi_d}(\widehat{f}_d) + \lambda_n \left\|\widehat{f}_d\right\|^2_{\mathcal{H}} - \inf_{f\in\mathcal{H}}\left(\lambda_n \|f\|^2_{\mathcal{H}} + R_{\phi_d}(f)\right) + \inf_{f\in\mathcal{H}}\left(\lambda_n \|f\|^2_{\mathcal{H}} + R_{\phi_d}(f) - R^*_{\phi_d}\right) \\ &= (I) + a(\lambda_n). \end{aligned}$$

By Theorem 3, we have

$$a(\lambda_n) \leqslant c_d(\sigma^p\lambda_n + C_\alpha(2d)^{\alpha p/2}\sigma^{-\alpha p}).$$

We will focus on how to bound (I) using the empirical process theory. Specifically, we will apply Lemma 6 in Bartlett et al. (2006). Because the $\widehat{f}_d$ is estimated using the cross-fitting procedure, we can assume that $\widehat{p}_{y,a|z}$ are estimated using an independent dataset. Denote the minimizer of $\inf_{f\in\mathcal{H}}\left(\lambda_n \|f\|^2_{\mathcal{H}} + R_{\phi_d}(f)\right)$ as f^*_{d,λ_n}. For the simplicity of the notations, we omit X in $U_+(\boldsymbol{X})$ and

$L_-(\boldsymbol{X})$ when there is no ambiguity. Let

$$\begin{aligned} L_f &= \Big\{\lambda_n \|f\|_{\mathcal{H}}^2 + U_+ \left\{\phi_d(f(\boldsymbol{X})) - \phi_d(f^*_{d,\lambda_n}(\boldsymbol{X}))\right\} + \\ &\quad L_- \left\{\phi_d(f(\boldsymbol{X})) - \phi_d(f^*_{d,\lambda_n}(\boldsymbol{X}))\right\} - \lambda \left\|f^*_{d,\lambda_n}\right\|_{\mathcal{H}}^2,\ f \in \mathcal{B}(B_n)\Big\}, \end{aligned}$$

where $\mathcal{B}(B_n)$ is defined in Lemma 3. By Lemma 3, we have

$$\|l_f\|_\infty \leqslant 2M + 2\left|U_+\phi_d(f(\boldsymbol{X})) + L_-\phi_d(f(\boldsymbol{X}))\right| = O(\lambda_n^{-1}),$$

for any $f \in \mathcal{B}(B_n)$. It can be shown that $E[l_f^2] \leqslant c'_n E[l_f]$, where $c'_n = O(\lambda_n^{-1})$ (see Zhao et al. (2012, 2019)). Suppose that l_f satisfies that $E_n[l_f] \leqslant \epsilon/2$ and $E[l_f] \geqslant \epsilon$ for some $\epsilon > 0$ to be chosen. By the continuity of ϕ_d, there exists an $f' = t'f + (1-t')f^*_{d,\lambda_n} \in \mathcal{B}(B_n)$, for some t', such that $E[l_{f'}] = \epsilon$. In addition, by the convexity of ϕ_d, we have

$$\begin{aligned} E_n[l_{f'}] &\leqslant \lambda_n t' \|f\|_{\mathcal{H}}^2 + t' E_n\left[U_+ \left\{\phi_d(f(\boldsymbol{X})) - \phi_d(f^*_{d,\lambda_n}(\boldsymbol{X}))\right\}\right] + \\ &\quad t' E_n\left[L_- \left\{\phi_d(f(\boldsymbol{X})) - \phi_d(f^*_{d,\lambda_n}(\boldsymbol{X}))\right\}\right] - t'\lambda_n \left\|f^*_{d,\lambda_n}\right\|_{\mathcal{H}}^2 \\ &\leqslant \epsilon/2. \end{aligned}$$

To apply Lemma 6 in Bartlett et al. (2006), we need to choose ϵ such that that

$$E_n[l_{\hat{f}_d}] \leqslant \epsilon/2.$$

To show this, notice that

$$\begin{aligned}
E_n[l_{\widehat{f}_d}] &= E_n\left[\lambda_n\left\|\widehat{f}_d\right\|_{\mathcal{H}}^2 + U_+\left\{\phi_d(\widehat{f}_d(\boldsymbol{X})) - \phi_d(f^*_{d,\lambda_n}(\boldsymbol{X}))\right\} + \right. \\
&\quad \left. L_-\left\{\phi_d(\widehat{f}_d(\boldsymbol{X})) - \phi_d(f^*_{d,\lambda_n}(\boldsymbol{X}))\right\} - \lambda_n\left\|f^*_{d,\lambda_n}\right\|_{\mathcal{H}}^2\right] \\
&= E_n\left[\lambda_n\left\|\widehat{f}_d\right\|_{\mathcal{H}}^2 + W_+\left\{\phi_d(\widehat{f}_d(\boldsymbol{X})) - \phi_d(f^*_{d,\lambda_n}(\boldsymbol{X}))\right\} + \right. \\
&\quad \left. W_-\left\{\phi_d(\widehat{f}_d(\boldsymbol{X})) - \phi_d(f^*_{d,\lambda_n}(\boldsymbol{X}))\right\} - \lambda_n\left\|f^*_{d,\lambda_n}\right\|_{\mathcal{H}}^2\right] + \\
&\quad E_n\left[\{U_+ - W_+\}\left\{\phi_d(\widehat{f}_d(\boldsymbol{X})) - \phi_d(f^*_{d,\lambda_n}(\boldsymbol{X}))\right\} + \right. \\
&\quad \left. \{L_- - W_-\}\left\{\phi_d(\widehat{f}_d(\boldsymbol{X})) - \phi_d(f^*_{d,\lambda_n}(\boldsymbol{X}))\right\}\right] \\
&\leqslant E_n\left[\{U_+ - W_+\}\left\{\phi_d(\widehat{f}_d(\boldsymbol{X})) - \phi_d(f^*_{d,\lambda_n}(\boldsymbol{X}))\right\} + \right. \\
&\quad \left. \{L_- - W_-\}\left\{\phi_d(\widehat{f}_d(\boldsymbol{X})) - \phi_d(f^*_{d,\lambda_n}(\boldsymbol{X}))\right\}\right].
\end{aligned}$$

By Lemma 5, we have that

$$E_n[l_{\widehat{f}_d}] \leqslant C\left\{n^{-1/2}\sigma_n^{(1-v/2)(1+\delta)p/2}\lambda_n^{-1/2} + n^{-(1+\gamma)\beta_p}\lambda_n^{-1/2} + n^{-\beta_\pi-\beta_p}\lambda_n^{-1/2}\right\},$$

where C denotes a generic constant. Therefore, Lemma 6 in Bartlett et al. (2006) implies that with probability tending to one,

$$E[l_{\widehat{f}_d}] \leqslant \epsilon,$$

provided that

$$\epsilon \geqslant \max\left\{\epsilon^*, \frac{Cc'_nK}{N}, \frac{CKB_n}{N}, Cn^{-1/2}\sigma_n^{(1-v/2)(1+\delta)p/2}\lambda_n^{-1/2} + Cn^{-(1+\gamma)\beta_p}\lambda_n^{-1/2} + Cn^{-\beta_\pi-\beta_p}\lambda_n^{-1/2}\right\},$$

where K is a constant, $B_n = O(\lambda_n^{-1/2})$, $c'_n = O(\lambda_n^{-1})$, and $\epsilon^* \geqslant 12\xi_{L_f}(\epsilon^*)$.

To choose ϵ^*, from Zhao et al. (2012), given a choice for B_n, for all $\sigma_n > 0$, $v \in (0, 2)$, $\delta > 0$, and

$\epsilon > 0$, we have

$$\sup_{P_n} \log N(B_n^{-1} L_f, \epsilon, L_2(P_n)) \leqslant C\sigma_n^{(1-v/2)(1+\delta)p} \epsilon^{-v}.$$

Thus, by Proposition 5.5 in Steinwart et al. (2007), we have that

$$\xi_{L_f}(\epsilon) \leqslant C \max \left\{ \epsilon^{1/2-v/4} \left(\frac{\sigma_n^{(1-v/2)(1+\delta)p} B_n^v}{N} \right)^{1/2}, \left(\frac{\sigma_n^{(1-v/2)(1+\delta)p} B_n^v}{N} \right)^{2/(2+v)} \right\}.$$

Thus, it suffices to choose

$$\epsilon^* = C\sigma_n^{(2-v)(1+\delta)p/(2+v)} \lambda^{-v/(v+2)} N^{-2/(2+v)}.$$

□

Lemma 3. *Suppose that* $n^{-1/2}\sigma_n^{(1-v/2)(1+\delta)p/2}\lambda_n^{-1/(2h)} \to 0$ *for some constant* $1 > h > 0$ *and* $\lambda_n \to 0$, *there exists a constant* M *such that*

$$\left\| \widehat{f}_d \right\|_{\mathcal{H}} \leqslant M\lambda_n^{-1/2},$$

and

$$\left\| f_{d,\lambda}^* \right\|_{\mathcal{H}} \leqslant M\lambda_n^{-1/2},$$

with a large probability, where f_{d,λ_n}^* *is the minimizer of* $\lambda_n \|f\|_{\mathcal{H}}^2 + R_{\phi_d}(f)$.

Proof. We first show that

$$\left\| f_{d,\lambda_n}^* \right\|_{\mathcal{H}} \leqslant M\lambda_n^{-1/2}.$$

By the definition of f^*_{d,λ_n}, we have that

$$\lambda \left\| f^*_{d,\lambda_n} \right\|^2_{\mathcal{H}} + R_{\phi_d}(f^*_{d,\lambda_n}) \leqslant R_{\phi_d}(0).$$

Since $R_{\phi_d}(f^*_{d,\lambda_n}) \geqslant 0$, we have that

$$\lambda_n \left\| f^*_{d,\lambda_n} \right\|^2_{\mathcal{H}} \leqslant R_{\phi_d}(0),$$

which is equivalent to $\left\| f^*_{d,\lambda_n} \right\|^2_{\mathcal{H}} \leqslant R^{1/2}_{\phi_d}(0)\lambda_n^{-1/2}$.

Next, we will show that

$$\left\| \widehat{f}_d \right\|_{\mathcal{H}} \leqslant M\lambda_n^{-1/2}$$

with probability approaching to 1. This result is non-trivial because for

$$\lambda_n \left\| f \right\|^2_{\mathcal{H}} + R_{\phi_d,n}(f),$$

where

$$R_{\phi_d,n}(f) = E_n \left[W_+ \phi_d(f(\boldsymbol{X})) + W_- \phi_d(-f(\boldsymbol{X})) \right],$$

we do not have the positivity for $R_{\phi_d,n}(f)$ due to possible negative weights, i.e., negative values of W_+ and W_-.

By the definition of $\widehat{f}_d$, we have that

$$\lambda_n \left\| \widehat{f}_d \right\|^2_{\mathcal{H}} + R_{\phi_d,n}(\widehat{f}_d) \leqslant R_{\phi_d,n}(0).$$

Notice that

$$\begin{aligned}\left|R_{\phi_d,n}(\widehat{f}_d)\right| &= \left|E_n\left[W_+\phi_d(\widehat{f}_d) + W_-\phi_d(-\widehat{f}_d)\right]\right| \\ &\leqslant CE_n\left[|\widehat{f}_d|\right]/d \\ &\leqslant C\left\|\widehat{f}_d\right\|_{\mathcal{H}}/d.\end{aligned}$$

since W_+ and W_- are bounded, and $\phi_d(\cdot)$ is d^{-1}-Lipschitz continuous. Thus, we have

$$\lambda\left\|\widehat{f}_d\right\|_{\mathcal{H}}^2 \leqslant R_{\phi_d,n}(0) + C\left\|\widehat{f}_d\right\|_{\mathcal{H}}/d,$$

which leads to $\left\|\widehat{f}_d\right\|_{\mathcal{H}} \leqslant M\lambda_n^{-1}$. This result can be strengthened by the following technique. Notice that

$$\lambda_n\left\|\widehat{f}_d\right\|_{\mathcal{H}}^2 \leqslant R_{\phi_d,n}(0) + \left|R_{\phi_d,n}(\widehat{f}_d) - R_{\phi_d}(\widehat{f}_d)\right|.$$

Let

$$\mathcal{F}_R = \{W_+\phi_d(f(\boldsymbol{X})) + W_-\phi_d(-f(\boldsymbol{X})), \text{ where } f \in \mathcal{B}(B_n)\},$$

where $B_n = M\lambda_n^{-1}$. From Zhao et al. (2012), we have that for all $\sigma_n > 0$, $v \in (0,2)$, $\delta > 0$, and $\epsilon > 0$, we have

$$\sup_{P_n} \log N(\mathcal{F}_R, \lambda_n^{-1}\epsilon, L_2(P_n)) \leqslant C\sigma_n^{(1-v/2)(1+\delta)p}\epsilon^{-v}.$$

Thus, by the maximal inequality in Lemma 19.38 in van der Vaart (2000), we have that

$$\left|R_{\phi_d,n}(\widehat{f}_d) - R_{\phi_d}(\widehat{f}_d)\right| \lesssim n^{-1/2}\sigma_n^{(1-v/2)(1+\delta)p/2}\lambda_n^{-1}.$$

Thus, we have that $\left\|\widehat{f}_d\right\|_{\mathcal{H}} \lesssim M\left\{n^{-1/2}\sigma_n^{(1-v/2)(1+\delta)p/2}\right\}^{1/2}\lambda_n^{-1}$. Denote $\kappa_n \equiv \left\{n^{-1/2}\sigma_n^{(1-v/2)(1+\delta)p/2}\right\}^{1/2}$. We can set $B_n = M\kappa_n\lambda_n^{-1}$, and repeat the same argument to conclude that

$$\left\|\widehat{f}_d\right\|_{\mathcal{H}} \lesssim M\left\{n^{-1/2}\sigma_n^{(1-v/2)(1+\delta)p/2}\right\}^{3/4}\lambda_n^{-1}.$$

Similarly, iteratively using this argument, we can conclude that for any constant $k \geqslant 1$, we have that

$$\left\|\widehat{f}_d\right\|_{\mathcal{H}} \lesssim M\left\{n^{-1/2}\sigma_n^{(1-v/2)(1+\delta)p/2}\right\}^{1-1/2^k}\lambda_n^{-1}.$$

Take k such that $\frac{2^k}{2^{k+1}-2} \leqslant 1/(2h)$, we have that

$$\begin{aligned}
&\left\|\widehat{f}_d\right\|_{\mathcal{H}} \\
\lesssim\ & M\left\{n^{-1/2}\sigma_n^{(1-v/2)(1+\delta)p/2}\lambda_n^{-2^k/(2^{k+1}-2)}\right\}^{1-1/2^k}\lambda_n^{-1/2} \\
=\ & M\left\{n^{-1/2}\sigma_n^{(1-v/2)(1+\delta)p/2}\lambda_n^{-1/(2h)}\right\}^{1-1/2^k}\lambda_n^{-1/2}.
\end{aligned}$$

By $n^{-1/2}\sigma_n^{(1-v/2)(1+\delta)p/2}\lambda_n^{-1/(2h)} \to 0$, we have that

$$\left\|\widehat{f}_d\right\|_{\mathcal{H}} \lesssim \lambda_n^{-1/2}.$$

□

Lemma 4. *Assuming* $\sup_x |\widehat{Q}_k(\boldsymbol{X}) - Q_k(\boldsymbol{X})| = O_p(n^{-\beta_p})$ *and* $\sup_x |\widehat{P}_k(\boldsymbol{X}) - P_k(\boldsymbol{X})| = O_p(n^{-\beta_p})$, *we have*

$$\begin{aligned}
\left|\min\left\{\widehat{Q}_1(\boldsymbol{X}),\cdots,\widehat{Q}_K(\boldsymbol{X})\right\} - \min\left\{Q_1(\boldsymbol{X}),\cdots,Q_K(\boldsymbol{X})\right\}\right| &\lesssim O_p(Kn^{-\beta_p}), \\
\left|\min\left\{\widehat{P}_1(\boldsymbol{X}),\cdots,\widehat{P}_K(\boldsymbol{X})\right\} - \min\left\{P_1(\boldsymbol{X}),\cdots,P_K(\boldsymbol{X})\right\}\right| &\lesssim O_p(Kn^{-\beta_p}).
\end{aligned}$$

Proof. We will only prove the minimum case. The maximum case can be proved likewise. First, we show that when $K = 2$, we have

$$\begin{aligned} & \left|\min\left\{\widehat{Q}_1(\boldsymbol{X}), \widehat{Q}_2(\boldsymbol{X})\right\} - \min\left\{Q_1(\boldsymbol{X}), Q_2(\boldsymbol{X})\right\}\right| \\ \leqslant\ & |\widehat{Q}_1(\boldsymbol{X}) - Q_1(\boldsymbol{X})| + |\widehat{Q}_2(\boldsymbol{X}) - Q_2(\boldsymbol{X})|. \end{aligned}$$

To show this, we directly calculate

$$\begin{aligned} & \left|\min\left\{\widehat{Q}_1(\boldsymbol{X}), \widehat{Q}_2(\boldsymbol{X})\right\} - \min\left\{Q_1(\boldsymbol{X}), Q_2(\boldsymbol{X})\right\}\right| \\ =\ & \left|\frac{\widehat{Q}_1(\boldsymbol{X}) + \widehat{Q}_2(\boldsymbol{X}) - |\widehat{Q}_1(\boldsymbol{X}) - \widehat{Q}_2(\boldsymbol{X})|}{2} - \frac{Q_1(\boldsymbol{X}) + Q_2(\boldsymbol{X}) - |Q_1(\boldsymbol{X}) - Q_2(\boldsymbol{X})|}{2}\right| \\ \leqslant\ & \frac{1}{2}\left|\widehat{Q}_1(\boldsymbol{X}) - Q_1(\boldsymbol{X})\right| + \frac{1}{2}\left|\widehat{Q}_2(\boldsymbol{X}) - Q_2(\boldsymbol{X})\right| + \frac{1}{2}\left|\left|Q_1(\boldsymbol{X}) - Q_2(\boldsymbol{X})\right| - \left|\widehat{Q}_1(\boldsymbol{X}) - \widehat{Q}_2(\boldsymbol{X})\right|\right| \\ \leqslant\ & \frac{1}{2}\left|\widehat{Q}_1(\boldsymbol{X}) - Q_1(\boldsymbol{X})\right| + \frac{1}{2}\left|\widehat{Q}_2(\boldsymbol{X}) - Q_2(\boldsymbol{X})\right| + \frac{1}{2}\left|Q_1(\boldsymbol{X}) - \widehat{Q}_1(\boldsymbol{X}) + \widehat{Q}_2(\boldsymbol{X}) - Q_2(\boldsymbol{X})\right| \\ \leqslant\ & |\widehat{Q}_1(\boldsymbol{X}) - Q_1(\boldsymbol{X})| + |\widehat{Q}_2(\boldsymbol{X}) - Q_2(\boldsymbol{X})|. \end{aligned}$$

For general K, we have

$$\begin{aligned} & \left|\min\left\{\widehat{Q}_1(\boldsymbol{X}), \cdots, \widehat{Q}_K(\boldsymbol{X})\right\} - \min\left\{Q_1(\boldsymbol{X}), \cdots, Q_K(\boldsymbol{X})\right\}\right| \\ =\ & \left|\min\left\{\widehat{Q}_1(\boldsymbol{X}), \min\left\{\widehat{Q}_2(\boldsymbol{X}), \cdots, \widehat{Q}_K(\boldsymbol{X})\right\}\right\} - \min\left\{Q_1(\boldsymbol{X}), \min\left\{Q_2(\boldsymbol{X}), \cdots, Q_K(\boldsymbol{X})\right\}\right\}\right| \\ \leqslant\ & |\widehat{Q}_1(\boldsymbol{X}) - Q_1(\boldsymbol{X})| + \left|\min\left\{\widehat{Q}_2(\boldsymbol{X}), \cdots, \widehat{Q}_K(\boldsymbol{X})\right\} - \min\left\{Q_2(\boldsymbol{X}), \cdots, Q_K(\boldsymbol{X})\right\}\right| \\ \leqslant\ & \sum_{k=1}^{K} |\widehat{Q}_k(\boldsymbol{X}) - Q_k(\boldsymbol{X})| = O_p(Kn^{-\beta_p}). \end{aligned}$$

□

Lemma 5. *Under Conditions in Theorem 7, we have*

$$\begin{aligned}
\left|E_n\left[\{U_+ - W_+\}\,\phi_d(\widehat{f}_d(\boldsymbol{X}))\right]\right| &= O_p(n^{-1/2}\sigma_n^{(1-v/2)(1+\delta)p/2}\lambda_n^{-1/2} + n^{-(1+\gamma)\beta_p}\lambda_n^{-1/2} + n^{-\beta_\pi-\beta_p}\lambda_n^{-1/2}),\\
\left|E_n\left[\{U_+ - W_+\}\,\phi_d(f^*_{d,\lambda_n}(\boldsymbol{X}))\right]\right| &= O_p(n^{-1/2}\sigma_n^{(1-v/2)(1+\delta)p/2}\lambda_n^{-1/2} + n^{-(1+\gamma)\beta_p}\lambda_n^{-1/2} + n^{-\beta_\pi-\beta_p}\lambda_n^{-1/2}),\\
\left|E_n\left[\{L_- - W_-\}\,\phi_d(\widehat{f}_d(\boldsymbol{X}))\right]\right| &= O_p(n^{-1/2}\sigma_n^{(1-v/2)(1+\delta)p/2}\lambda_n^{-1/2} + n^{-(1+\gamma)\beta_p}\lambda_n^{-1/2} + n^{-\beta_\pi-\beta_p}\lambda_n^{-1/2}),\\
\left|E_n\left[\{L_- - W_-\}\,\phi_d(f^*_{d,\lambda_n}(\boldsymbol{X}))\right]\right| &= O_p(n^{-1/2}\sigma_n^{(1-v/2)(1+\delta)p/2}\lambda_n^{-1/2} + n^{-(1+\gamma)\beta_p}\lambda_n^{-1/2} + n^{-\beta_\pi-\beta_p}\lambda_n^{-1/2}).
\end{aligned}$$

Proof. For simplicity, we will only show

$$\left|E_n\left[\{U_+ - W_+\}\,\phi_d(\widehat{f}_d(\boldsymbol{X}))\right]\right| = O_p(n^{-1/2}\sigma_n^{(1-v/2)(1+\delta)p/2}\lambda_n^{-1/2} + n^{-(1+\gamma)\beta_p}\lambda_n^{-1/2} + n^{-\beta_\pi-\beta_p}\lambda_n^{-1/2}).$$

Notice that

$$\begin{aligned}
&\left|E_n\left[\{U_+ - W_+\}\,\phi_d(\widehat{f}_d(\boldsymbol{X}))\right]\right|\\
=\ &\left|E_n\left[\left\{U_+ - \widehat{U}_+ + \sum_{y,a,z}\widehat{B}_+(X,Y,A,Z;y,a,z)\right\}\phi_d(\widehat{f}_d(\boldsymbol{X}))\right]\right|\\
\leqslant\ &\left|E_n\left[\left\{U_+ - \widehat{U}_+ + \sum_{y,a,z}E\left[\widehat{B}_+(X,Y,A,Z;y,a,z)\mid X\right]\right\}\phi_d(\widehat{f}_d(\boldsymbol{X}))\right]\right|\\
&+\left|E_n\left[\left\{\widehat{B}_+(X,Y,A,Z;y,a,z) - \sum_{y,a,z}E\left[\widehat{B}_+(X,Y,A,Z;y,a,z)\mid X\right]\right\}\phi_d(\widehat{f}_d(\boldsymbol{X}))\right]\right|\\
=\ &I_1 + I_2.
\end{aligned}$$

We first consider I_2. Notice that

$$\widehat{B}_+(X,Y,A,Z;y,a,z) = \partial_{y,a,z}W_+(X;\widehat{Q}_1,\cdots,\widehat{Q}_K)\frac{1\{Z=z\}}{\widehat{\pi}_z(\boldsymbol{X})}(\widehat{p}_{y,a|z}(\boldsymbol{X}) - 1\{Y=y,A=a\}).$$

Thus, we have $\widehat{B}_+(X,Y,A,Z;y,a,z)$ can be bounded by a sufficient;y large constant C. Let

$$\mathcal{F}_{B,2} = \left\{ \left\{ \widehat{B}_+(X,Y,A,Z;y,a,z) - \sum_{y,a,z} E\left[\widehat{B}_+(X,Y,A,Z;y,a,z) \mid X\right] \right\} \phi_d(\widehat{f}_d(\boldsymbol{X})),\ f \in \mathcal{B}(B_n) \right\}.$$

The envelop function is given by $F_{B,2} = 16C\lambda_n^{-1/2}$. From Zhao et al. (2012), we have that for all $\sigma_n > 0$, $v \in (0,2)$, $\delta > 0$, and $\epsilon > 0$, we have

$$\sup_{P_n} \log N(\mathcal{F}_{B,2}, \lambda_n^{-1/2}\epsilon, L_2(P_n)) \leqslant C\sigma_n^{(1-v/2)(1+\delta)p}\epsilon^{-v}.$$

Thus, by the maximal inequality in Lemma 19.38 in van der Vaart (2000), we have that

$$I_2 \lesssim n^{-1/2}\sigma_n^{(1-v/2)(1+\delta)p/2}\lambda_n^{-1/2}.$$

Now, we consider I_1. Notice that

$$\begin{aligned}
& E\left[\widehat{B}_+(X,Y,A,Z;y,a,z) \mid X\right] \\
= \ & \partial_{y,a,z}W_+(X;\widehat{Q}_1,\cdots,\widehat{Q}_K)\frac{\pi_z(\boldsymbol{X})}{\widehat{\pi}_z(\boldsymbol{X})}(\widehat{p}_{y,a|z}(\boldsymbol{X}) - p_{y,a|z}(\boldsymbol{X})) \\
= \ & \partial_{y,a,z}W_+(X;\widehat{Q}_1,\cdots,\widehat{Q}_K)(\widehat{p}_{y,a|z}(\boldsymbol{X}) - p_{y,a|z}(\boldsymbol{X})) \\
& + \partial_{y,a,z}W_+(X;\widehat{Q}_1,\cdots,\widehat{Q}_K)\left\{\frac{\pi_z(\boldsymbol{X})}{\widehat{\pi}_z(\boldsymbol{X})} - 1\right\}(\widehat{p}_{y,a|z}(\boldsymbol{X}) - p_{y,a|z}(\boldsymbol{X}))
\end{aligned}$$

We have

$$\begin{aligned}
& I_1 \\
\leqslant \ & \left| E_n\left[\left\{U_+ - \widehat{U}_+ + \sum_{y,a,z}\partial_{y,a,z}W_+(X;\widehat{Q}_1,\cdots,\widehat{Q}_K)(\widehat{p}_{y,a|z}(\boldsymbol{X}) - p_{y,a|z}(\boldsymbol{X}))\right\}\phi_d(\widehat{f}_d(\boldsymbol{X}))\right]\right| + \\
& \left| E_n\left[\left\{\sum_{y,a,z}\partial_{y,a,z}W_+(X;\widehat{Q}_1,\cdots,\widehat{Q}_K)\left\{\frac{\pi_z(\boldsymbol{X})}{\widehat{\pi}_z(\boldsymbol{X})} - 1\right\}(\widehat{p}_{y,a|z}(\boldsymbol{X}) - p_{y,a|z}(\boldsymbol{X}))\right\}\phi_d(\widehat{f}_d(\boldsymbol{X}))\right]\right| \\
= \ & I_{11} + I_{12}.
\end{aligned}$$

For I_{12}, by Assumption C2), we have that $I_{12} \lesssim n^{-\beta_\pi - \beta_p} \lambda_n^{-1/2}$. To bound I_{11}, we have

$$
\begin{aligned}
& I_{11} \\
\leqslant & \left| (E_n - E) \left[\left\{ U_+ - \widehat{U}_+ + \sum_{y,a,z} \partial_{y,a,z} W_+(X; \widehat{Q}_1, \cdots, \widehat{Q}_K)(\widehat{p}_{y,a|z}(\boldsymbol{X}) - p_{y,a|z}(\boldsymbol{X})) \right\} \phi_d(\widehat{f}_d(\boldsymbol{X})) \right] \right| + \\
& \left| E \left[\left\{ U_+ - \widehat{U}_+ + \sum_{y,a,z} \partial_{y,a,z} W_+(X; \widehat{Q}_1, \cdots, \widehat{Q}_K)(\widehat{p}_{y,a|z}(\boldsymbol{X}) - p_{y,a|z}(\boldsymbol{X})) \right\} \phi_d(\widehat{f}_d(\boldsymbol{X})) \right] \right|.
\end{aligned}
$$

To bound the first term, we notice that

$$
\left| U_+ - \widehat{U}_+ + \sum_{y,a,z} \partial_{y,a,z} W_+(X; \widehat{Q}_1, \cdots, \widehat{Q}_K)(\widehat{p}_{y,a|z}(\boldsymbol{X}) - p_{y,a|z}(\boldsymbol{X})) \right| \leqslant C n^{-\beta_p}.
$$

Consider

$$
\mathcal{F}_{U,2} = \left\{ \left\{ U_+ - \widehat{U}_+ + \sum_{y,a,z} \partial_{y,a,z} W_+(X; \widehat{Q}_1, \cdots, \widehat{Q}_K)(\widehat{p}_{y,a|z}(\boldsymbol{X}) - p_{y,a|z}(\boldsymbol{X})) \right\} \phi_d(f_d(\boldsymbol{X})),\ f \in \mathcal{B}(B_n) \right\}.
$$

The envelop function is given by $F_{U,2} = C n^{-\beta_p} \lambda_n^{-1/2}$. Again, from Zhao et al. (2012), we have that for all $\sigma_n > 0$, $v \in (0, 2)$, $\delta > 0$, and $\epsilon > 0$, we have

$$
\sup_{P_n} \log N(\mathcal{F}_{B,2}, \lambda_n^{-1/2} \epsilon, L_2(P_n)) \leqslant C \sigma_n^{(1-v/2)(1+\delta)p} \epsilon^{-v}.
$$

Thus, by the maximal inequality in Lemma 19.38 in van der Vaart (2000), we have that

$$
\begin{aligned}
& \left| (E_n - E) \left[\left\{ U_+ - \widehat{U}_+ + \sum_{y,a,z} \partial_{y,a,z} W_+(X; \widehat{Q}_1, \cdots, \widehat{Q}_K)(\widehat{p}_{y,a|z}(\boldsymbol{X}) - p_{y,a|z}(\boldsymbol{X})) \right\} \phi_d(\widehat{f}_d(\boldsymbol{X})) \right] \right| \\
\lesssim & \ n^{-\beta_p - 1/2} \sigma_n^{(1-v/2)(1+\delta)p/2} \lambda_n^{-1/2}.
\end{aligned}
$$

Thus, to bound I_1. we only need to bound

$$\left| E\left[\left\{ U_+ - \widehat{U}_+ + \sum_{y,a,z} \partial_{y,a,z} W_+(X; \widehat{Q}_1, \cdots, \widehat{Q}_K)(\widehat{p}_{y,a|z}(\boldsymbol{X}) - p_{y,a|z}(\boldsymbol{X})) \right\} \phi_d(\widehat{f}_d(\boldsymbol{X})) \right] \right|.$$

We write

$$\begin{aligned}
& U_+ - \widehat{U}_+ + \sum_{y,a,z} \partial_{y,a,z} W_+(X; \widehat{Q}_1, \cdots, \widehat{Q}_K)(\widehat{p}_{y,a|z}(\boldsymbol{X}) - p_{y,a|z}(\boldsymbol{X})) \\
= \; & |\min\{Q_1(\boldsymbol{X}), \cdots, Q_K(\boldsymbol{X})\}| \, 1\{\min\{Q_1(\boldsymbol{X}), \cdots, Q_K(\boldsymbol{X})\} \leqslant 0\} \\
& - \left| \min\left\{ \widehat{Q}_1(\boldsymbol{X}), \cdots, \widehat{Q}_K(\boldsymbol{X}) \right\} \right| 1\left\{ \min\left\{ \widehat{Q}_1(\boldsymbol{X}), \cdots, \widehat{Q}_K(\boldsymbol{X}) \right\} \leqslant 0 \right\} \\
& - 1\left\{ \min\left\{ \widehat{Q}_1(\boldsymbol{X}), \cdots, \widehat{Q}_K(\boldsymbol{X}) \right\} \leqslant 0 \right\} \sum_{k=1}^{K} 1\left\{ \widehat{Q}_k(\boldsymbol{X}) \leqslant \min_{k' \neq k} \widehat{Q}_{k'}(\boldsymbol{X}) \right\} \\
& \times \sum_{y,a,z} \partial_{y,a,z} Q_k(\boldsymbol{X})(\widehat{p}_{y,a|z}(\boldsymbol{X}) - p_{y,a|z}(\boldsymbol{X})).
\end{aligned}$$

We define the following events

$$\begin{aligned}
\Omega &= \{\min\{Q_1(\boldsymbol{X}), \cdots, Q_K(\boldsymbol{X})\} \leqslant 0\}, \\
\widehat{\Omega} &= \left\{ \min\left\{ \widehat{Q}_1(\boldsymbol{X}), \cdots, \widehat{Q}_K(\boldsymbol{X}) \right\} \leqslant 0 \right\}, \\
\Omega_k &= \left\{ Q_k(\boldsymbol{X}) - \min_{k' \neq k} Q_{k'}(\boldsymbol{X}) \leqslant 0 \right\}, \\
\widehat{\Omega}_k &= \left\{ \widehat{Q}_k(\boldsymbol{X}) - \min_{k' \neq k} \widehat{Q}_{k'}(\boldsymbol{X}) \leqslant 0 \right\}.
\end{aligned}$$

On the event $\Omega \cap \widehat{\Omega} \cap \Omega_k \cap \widehat{\Omega}_k$, we have

$$\begin{aligned}
& U_+ - \widehat{U}_+ + \sum_{y,a,z} \partial_{y,a,z} W_+(X; \widehat{Q}_1, \cdots, \widehat{Q}_K)(\widehat{p}_{y,a|z}(\boldsymbol{X}) - p_{y,a|z}(\boldsymbol{X})) \\
= \; & -Q_k(\boldsymbol{X}) + \widehat{Q}_k(\boldsymbol{X}) - \sum_{y,a,z} \partial_{y,a,z} Q_k(\boldsymbol{X})(\widehat{p}_{y,a|z}(\boldsymbol{X}) - p_{y,a|z}(\boldsymbol{X})) \\
= \; & 0.
\end{aligned}$$

Thus, we just need to consider the event $\left\{ \Omega \cap \widehat{\Omega} \cap \{\cup_k (\Omega_k \cap \widehat{\Omega}_k)\} \right\}^c = (\Omega \cap \widehat{\Omega})^c \cup \{\cup_k (\Omega_k \cap \widehat{\Omega}_k)\}^c$.

On event $(\Omega \cap \widehat{\Omega})^c = \Omega^c \cup \widehat{\Omega}^c$, we have

$$
\begin{aligned}
& \left| U_+ - \widehat{U}_+ + \sum_{y,a,z} \partial_{y,a,z} W_+(X; \widehat{Q}_1, \cdots, \widehat{Q}_K)(\widehat{p}_{y,a|z}(\boldsymbol{X}) - p_{y,a|z}(\boldsymbol{X})) \right| 1\{\Omega^c \cup \widehat{\Omega}^c\} \\
\leqslant & \left| U_+ - \widehat{U}_+ + \sum_{y,a,z} \partial_{y,a,z} W_+(X; \widehat{Q}_1, \cdots, \widehat{Q}_K)(\widehat{p}_{y,a|z}(\boldsymbol{X}) - p_{y,a|z}(\boldsymbol{X})) \right| \\
& \times \left\{ 1\left\{ \min\left\{ \widehat{Q}_1(\boldsymbol{X}), \cdots, \widehat{Q}_K(\boldsymbol{X}) \right\} > 0 \right\} + 1\left\{ \min\left\{ Q_1(\boldsymbol{X}), \cdots, Q_K(\boldsymbol{X}) \right\} > 0 \right\} \right\} \\
= & \left| |\min\left\{ Q_1(\boldsymbol{X}), \cdots, Q_K(\boldsymbol{X}) \right\}| 1\left\{ \min\left\{ Q_1(\boldsymbol{X}), \cdots, Q_K(\boldsymbol{X}) \right\} \leqslant 0,\ \min\left\{ \widehat{Q}_1(\boldsymbol{X}), \cdots, \widehat{Q}_K(\boldsymbol{X}) \right\} > 0 \right\} \right. \\
& - \left| \min\left\{ \widehat{Q}_1(\boldsymbol{X}), \cdots, \widehat{Q}_K(\boldsymbol{X}) \right\} \right| 1\left\{ \min\left\{ \widehat{Q}_1(\boldsymbol{X}), \cdots, \widehat{Q}_K(\boldsymbol{X}) \right\} \leqslant 0,\ \min\left\{ Q_1(\boldsymbol{X}), \cdots, Q_K(\boldsymbol{X}) \right\} > 0 \right\} \\
& -1\left\{ \min\left\{ \widehat{Q}_1(\boldsymbol{X}), \cdots, \widehat{Q}_K(\boldsymbol{X}) \right\} \leqslant 0,\ \min\left\{ Q_1(\boldsymbol{X}), \cdots, Q_K(\boldsymbol{X}) \right\} > 0 \right\} \sum_{k=1}^{K} 1\left\{ \widehat{Q}_k(\boldsymbol{X}) \leqslant \min_{k' \neq k} \widehat{Q}_{k'}(\boldsymbol{X}) \right\} \\
& \left. \times \sum_{y,a,z} \partial_{y,a,z} Q_k(\boldsymbol{X})(\widehat{p}_{y,a|z}(\boldsymbol{X}) - p_{y,a|z}(\boldsymbol{X})) \right| \\
= & \left| |\min\left\{ Q_1(\boldsymbol{X}), \cdots, Q_K(\boldsymbol{X}) \right\}| 1\left\{ \min\left\{ Q_1(\boldsymbol{X}), \cdots, Q_K(\boldsymbol{X}) \right\} < 0,\ \min\left\{ \widehat{Q}_1(\boldsymbol{X}), \cdots, \widehat{Q}_K(\boldsymbol{X}) \right\} > 0 \right\} \right. \\
& - \left| \min\left\{ \widehat{Q}_1(\boldsymbol{X}), \cdots, \widehat{Q}_K(\boldsymbol{X}) \right\} \right| 1\left\{ \min\left\{ \widehat{Q}_1(\boldsymbol{X}), \cdots, \widehat{Q}_K(\boldsymbol{X}) \right\} \leqslant 0,\ \min\left\{ Q_1(\boldsymbol{X}), \cdots, Q_K(\boldsymbol{X}) \right\} > 0 \right\} \\
& -1\left\{ \min\left\{ \widehat{Q}_1(\boldsymbol{X}), \cdots, \widehat{Q}_K(\boldsymbol{X}) \right\} \leqslant 0,\ \min\left\{ Q_1(\boldsymbol{X}), \cdots, Q_K(\boldsymbol{X}) \right\} > 0 \right\} \sum_{k=1}^{K} 1\left\{ \widehat{Q}_k(\boldsymbol{X}) \leqslant \min_{k' \neq k} \widehat{Q}_{k'}(\boldsymbol{X}) \right\} \\
& \left. \times \sum_{y,a,z} \partial_{y,a,z} Q_k(\boldsymbol{X})(\widehat{p}_{y,a|z}(\boldsymbol{X}) - p_{y,a|z}(\boldsymbol{X})) \right| \\
\leqslant & \left| \min\left\{ Q_1(\boldsymbol{X}), \cdots, Q_K(\boldsymbol{X}) \right\} - \min\left\{ \widehat{Q}_1(\boldsymbol{X}), \cdots, \widehat{Q}_K(\boldsymbol{X}) \right\} \right| \\
& \times \left\{ 1\left\{ \min\left\{ Q_1(\boldsymbol{X}), \cdots, Q_K(\boldsymbol{X}) \right\} < 0,\ \min\left\{ \widehat{Q}_1(\boldsymbol{X}), \cdots, \widehat{Q}_K(\boldsymbol{X}) \right\} > 0 \right\} \right. \\
& \left. +1\left\{ \min\left\{ \widehat{Q}_1(\boldsymbol{X}), \cdots, \widehat{Q}_K(\boldsymbol{X}) \right\} \leqslant 0,\ \min\left\{ Q_1(\boldsymbol{X}), \cdots, Q_K(\boldsymbol{X}) \right\} > 0 \right\} \right\} \\
& +1\left\{ \min\left\{ \widehat{Q}_1(\boldsymbol{X}), \cdots, \widehat{Q}_K(\boldsymbol{X}) \right\} \leqslant 0,\ \min\left\{ Q_1(\boldsymbol{X}), \cdots, Q_K(\boldsymbol{X}) \right\} > 0 \right\} \sum_{k=1}^{K} 1\left\{ \widehat{Q}_k(\boldsymbol{X}) \leqslant \min_{k' \neq k} \widehat{Q}_{k'}(\boldsymbol{X}) \right\} \\
& \times \sum_{y,a,z} |\partial_{y,a,z} Q_k(\boldsymbol{X})| \left(\widehat{p}_{y,a|z}(\boldsymbol{X}) - p_{y,a|z}(\boldsymbol{X})\right) \\
\leqslant & CKn^{-\beta_p} \left\{ 1\left\{ \min\left\{ Q_1(\boldsymbol{X}), \cdots, Q_K(\boldsymbol{X}) \right\} < 0,\ \min\left\{ \widehat{Q}_1(\boldsymbol{X}), \cdots, \widehat{Q}_K(\boldsymbol{X}) \right\} > 0 \right\} \right. \\
& \left. +1\left\{ \min\left\{ \widehat{Q}_1(\boldsymbol{X}), \cdots, \widehat{Q}_K(\boldsymbol{X}) \right\} \leqslant 0,\ \min\left\{ Q_1(\boldsymbol{X}), \cdots, Q_K(\boldsymbol{X}) \right\} > 0 \right\} \right\} \\
\leqslant & CKn^{-\beta_p} 1\left\{ 0 \neq |\min\left\{ Q_1(\boldsymbol{X}), \cdots, Q_K(\boldsymbol{X}) \right\}| \leqslant \left| \min\left\{ \widehat{Q}_1(\boldsymbol{X}), \cdots, \widehat{Q}_K(\boldsymbol{X}) \right\} - \min\left\{ Q_1(\boldsymbol{X}), \cdots, Q_K(\boldsymbol{X}) \right\} \right| \right\}.
\end{aligned}
$$

From the inequalities above, we have that

$$
\begin{aligned}
&\left|E\left[\left\{U_+ - \widehat{U}_+ + \sum_{y,a,z} \partial_{y,a,z} W_+(X; \widehat{Q}_1, \cdots, \widehat{Q}_K)(\widehat{p}_{y,a|z}(\boldsymbol{X}) - p_{y,a|z}(\boldsymbol{X}))\right\} \phi_d(\widehat{f}_d(\boldsymbol{X})) 1\{\Omega^c \cup \widehat{\Omega}^c\}\right]\right| \\
\leqslant\ & CKn^{-\beta_p}\lambda_n^{-1/2} P\left\{0 \neq |\min\left\{Q_1(\boldsymbol{X}), \cdots, Q_K(\boldsymbol{X})\right\}| \leqslant\right. \\
& \left.\left|\min\left\{\widehat{Q}_1(\boldsymbol{X}), \cdots, \widehat{Q}_K(\boldsymbol{X})\right\} - \min\left\{Q_1(\boldsymbol{X}), \cdots, Q_K(\boldsymbol{X})\right\}\right|\right\} \\
\leqslant\ & CKn^{-\beta_p}\lambda_n^{-1/2} n^{-\alpha\gamma}.
\end{aligned}
$$

On event $\{\cup_k(\Omega_k \cap \widehat{\Omega}_k)\}^c \cap (\Omega \cap \widehat{\Omega})$, we have that

$$
\begin{aligned}
&\left|U_+ - \widehat{U}_+ + \sum_{y,a,z} \partial_{y,a,z} W_+(X; \widehat{Q}_1, \cdots, \widehat{Q}_K)(\widehat{p}_{y,a|z}(\boldsymbol{X}) - p_{y,a|z}(\boldsymbol{X}))\right| 1\{\{\cup_k(\Omega_k \cap \widehat{\Omega}_k)\}^c \cap (\Omega \cap \widehat{\Omega})\} \\
\leqslant\ & CKn^{-\beta_p} 1\{\{\cup_k(\Omega_k \cap \widehat{\Omega}_k)\}^c \cap (\Omega \cap \widehat{\Omega})\}.
\end{aligned}
$$

Thus, we have

$$
\begin{aligned}
&\left|E\left[\left\{U_+ - \widehat{U}_+ + \sum_{y,a,z} \partial_{y,a,z} W_+(X; \widehat{Q}_1, \cdots, \widehat{Q}_K)(\widehat{p}_{y,a|z}(\boldsymbol{X}) - p_{y,a|z}(\boldsymbol{X}))\right\} \phi_d(\widehat{f}_d(\boldsymbol{X})) 1\{\Omega^c \cup \widehat{\Omega}^c\}\right]\right| \\
\leqslant\ & CKn^{-\beta_p}\lambda_n^{-1/2} P\left\{\{\cup_k(\Omega_k \cap \widehat{\Omega}_k)\}^c\right\} \\
\leqslant\ & CKn^{-\beta_p}\lambda_n^{-1/2} P\left\{\text{there exist some } k \text{ and } k' \text{ such that}\right. \\
& \left.\widehat{Q}_k \leqslant \widehat{Q}_{k'} \text{ and } Q_k > Q_{k'}\right\} \\
\leqslant\ & CKn^{-\beta_p}\lambda_n^{-1/2} \sum_{k,k'} P\left\{0 < |Q_k - Q_{k'}| \leqslant |Q_k - Q_{k'} - \widehat{Q}_k + \widehat{Q}_{k'}|\right\} \\
\leqslant\ & CK^3 n^{-\beta_p}\lambda_n^{-1/2} n^{-\beta_p\gamma}.
\end{aligned}
$$

This concludes the proof. □

4 Additional simulation results

4.1 Performance of the proposed framework with a deferred option

In this section, we provide more comprehensive performance evaluation of the proposed methods. In addition to the metrics mentioned in the main text, we also have added the calculation of two new metrics for deferral: 1) the proportion of deferred subjects with treatment-effect bounds crossing zero ("PropCorrectDef"); and 2) the proportion of incorrect treatment assignment among non-deferred subjects ("PropIncorrectTrt").

Results for Settings $(2,1)$ and $(2,2)$ are summarized in Figures 2 and 3. When $d = 0.5$, no one will be deferred, and thus, the proportion of deferred subjects with treatment-effect bounds crossing zero cannot be calculated. For all settings and d's, the proposed methods achieve higher proportion of deferred subjects with treatment-effect bounds crossing zero and lower proportion of incorrect treatment assignment among non-deferred subjects, compared with the Wald method. Comparing Augmented ERM with the traditional ERM, the proposed Augmented ERM performs better than ERM for most of the cases. But when d is small in Setting (1,2), although the traditional ERM achieves higher proportion of deferred subjects with treatment-effect bounds crossing zero, Augmented ERM has lower proportion of incorrect treatment assignment among non-deferred subjects, which leads to higher value on the non-deferred group.

Figures 4 and 5 provide a summary of the comparison metrics for each method in Settings $(1,1)$ and $(2,1)$ across 200 repetitions. In Settings $(1,1)$ and $(2,1)$, the main effects are generated from a linear model; in Settings $(1,2)$ and $(2,2)$, the main effects are generated from a non-linear model. It can be observed that the performance of the proposed method vs. others is similar to those in Settings $(1,2)$ and $(2,2)$. The proposed A-ERM yields ITRs with the smallest loss in value and the highest average outcomes on the non-deferred subgroup. These show that our proposed framework achieves robust performance gain when compared with other methods under different specification of the main effects.

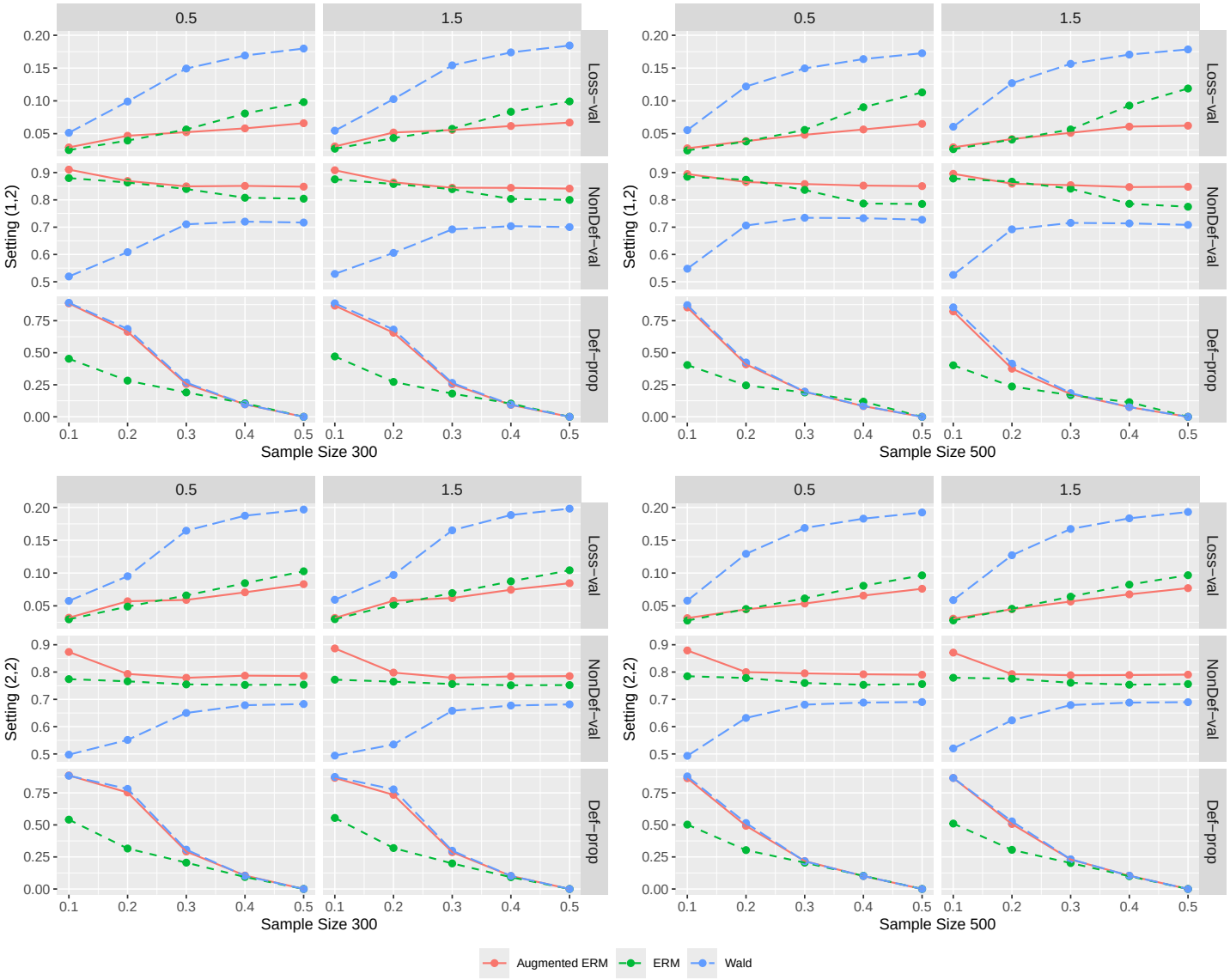


Figure 2: Simulation results of Settings $(2,1)$ and $(2,2)$ on overall performance over 200 repeats.

4.2 Performance of the proposed framework under varying IV strength

In this section, we evaluate the performance of the proposed methods under varying IV strength.

To generate the data, we fixed the data-generating process as $(2,2)$ and varied the effect of Z on the

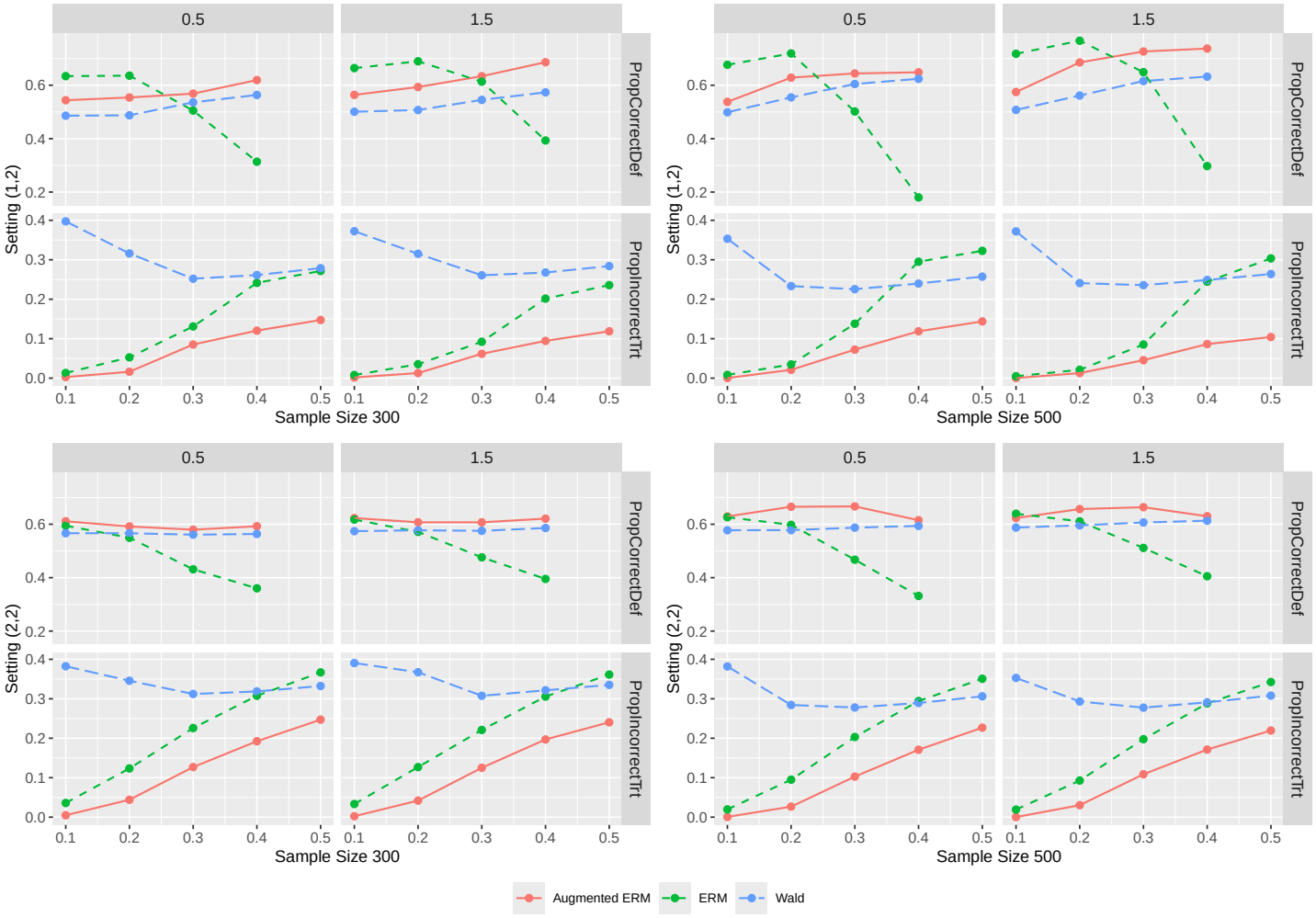


Figure 3: Simulation results of Settings $(2,1)$ and $(2,2)$ on metrics for deferral over 200 repeats.

treatment A from weak to strong. Specifically, we generate the data following:

$$\log\left\{\frac{P(A=1\mid \boldsymbol{X},U,Z)}{P(A=-1\mid \boldsymbol{X},U,Z)}\right\}=\zeta Z+X_1-7X_2+\xi(1+X_1)U,$$

and

$$\log\left\{\frac{P(Y_a=1\mid \boldsymbol{X},U,Z)}{P(Y_a=-1\mid \boldsymbol{X},U,Z)}\right\}=g_1(\boldsymbol{X},U)+g_2(\boldsymbol{X},U,a),$$

where $g_1(\boldsymbol{X},U)=1-X_1^2+X_2^2+\xi X_1X_2U$; $g_2(\boldsymbol{X},U,a)=a(4+X_1-0.25\xi X_1^2-2\times 1\{X_2\leqslant 0\})$. The parameter ζ controls the strength of IV: strong - 12, existing - 8, and weak - 4.

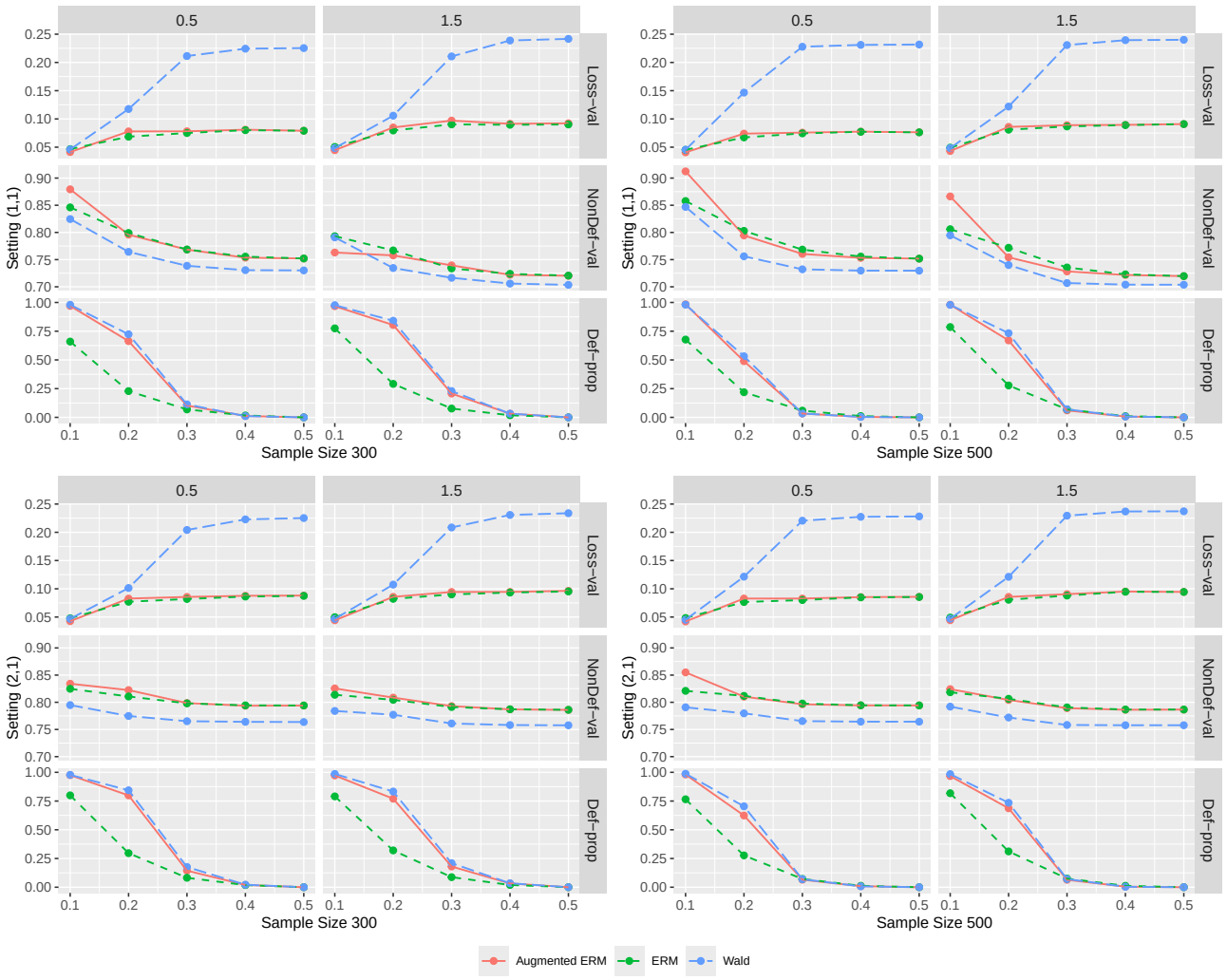


Figure 4: Simulation results for Settings $(1,1)$ and $(2,1)$ over 200 repeats.

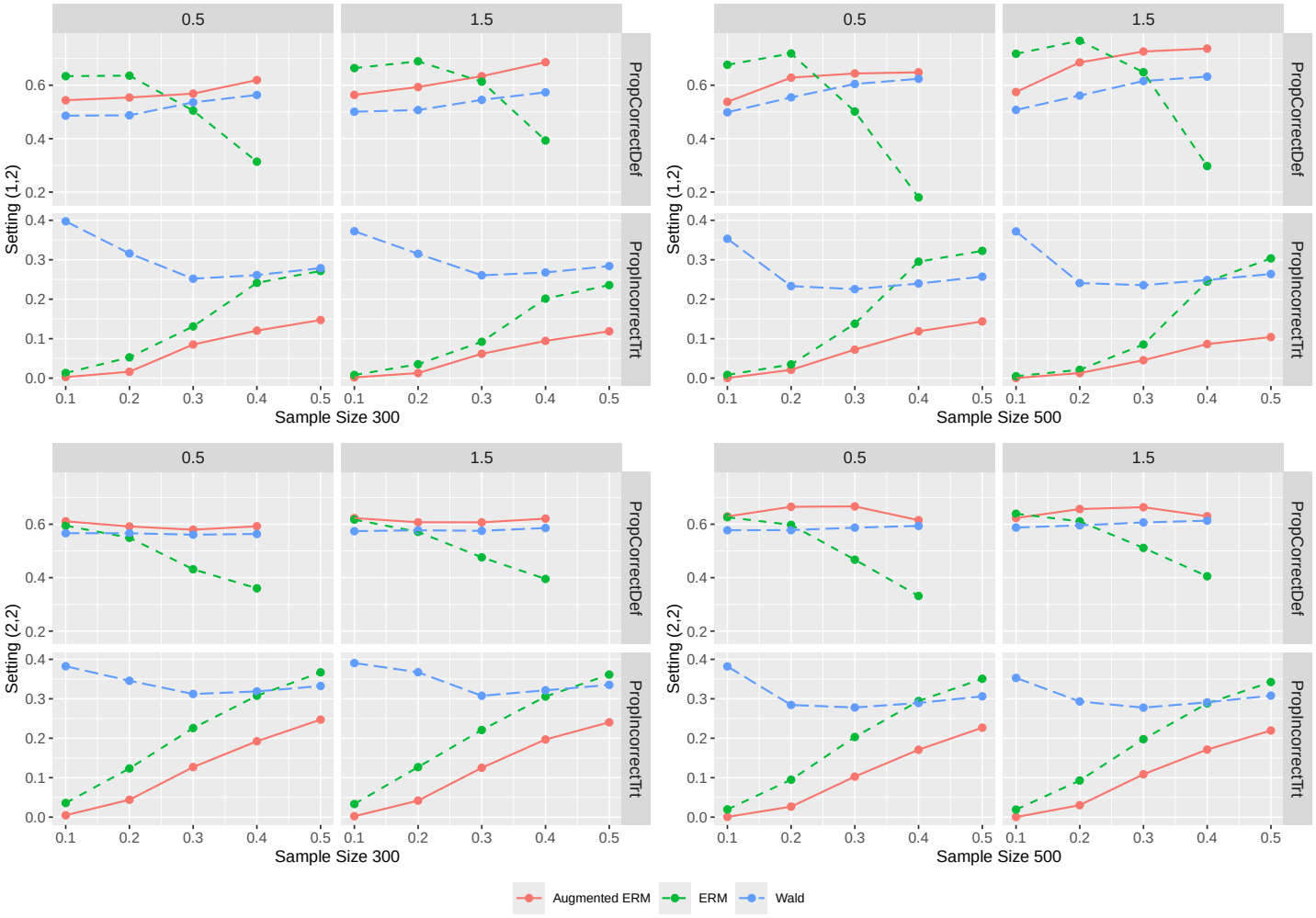


Figure 5: Simulation results of Settings $(1, 1)$ and $(2, 1)$ on metrics for deferral over 200 repeats.

Results are summarized in Figure 6. Overall, the proposed methods achieve minimal loss in value and highest value on non-deferred patients; they also achieve relative high proportion of correct deferring among deferred patients and lowest error rate of wrong treatment assignment among non-deferred patients. As the instrument becomes stronger, the performance gap between Augmented ERM and the traditional ERM widens.

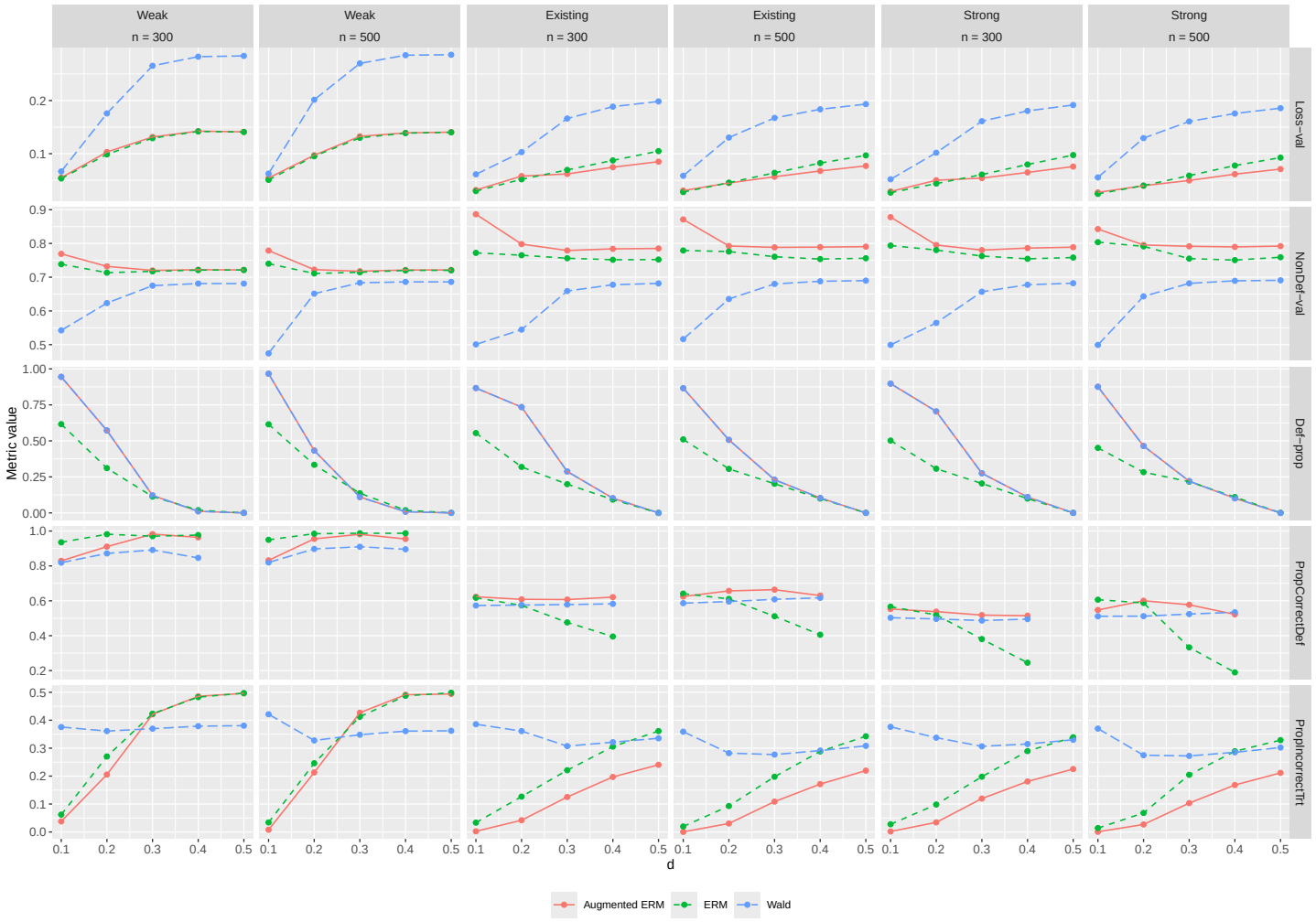


Figure 6: Simulation results under varying IV strength over 200 repeats.

4.3 Performance of the proposed framework under varying number of covariates

In this section, we evaluate the performance of the proposed methods under varying number of covariates. To generate the data, we fixed the data-generating process as $(2, 2)$ and varied the number of covariates. Specifically, we generate the data following:

$$\log\left\{\frac{P(A=1 \mid \boldsymbol{X}, U, Z)}{P(A=-1 \mid \boldsymbol{X}, U, Z)}\right\} = 8Z + X_1 - 7X_2 + \xi(1+X_1)U,$$

and

$$\log\left\{\frac{P(Y_a=1\mid \boldsymbol{X},U,Z)}{P(Y_a=-1\mid \boldsymbol{X},U,Z)}\right\}=g_1(\boldsymbol{X},U)+g_2(\boldsymbol{X},U,a),$$

where $g_1(\boldsymbol{X},U)=1-X_1^2+X_2^2+\xi X_1X_2U$; $g_2(\boldsymbol{X},U,a)=a(4+X_1-0.25\xi X_1^2-2\times 1\{X_2\leqslant 0\})$. Further, we generate 5 or 10 covariates in total to evaluate the performance under more covariates. Results are summarized in Figure 7. Overall, the simulation with $p=10$ is more difficult than settings with $p=5$, especially for our proposed methods. The Augmented ERM outperforms the traditional ERM in all settings; when $p=10$, the Wald method seems have higher proportion of correctly identifying deferred patients but with a much higher error rate of incorrect treatment assignment.

4.4 Performance of the proposed framework under point-identifiable settings

In this section, we consider a new simulation setting, where the CATE can be identified and the Wald estimator is consistent. Specifically, we generate the potential outcomes following Setting (2,2), i.e.,

$$\log\left\{\frac{P(Y_a=1\mid \boldsymbol{X},U,Z)}{P(Y_a=-1\mid \boldsymbol{X},U,Z)}\right\}=g_1(\boldsymbol{X},U)+g_2(\boldsymbol{X},U,a),$$

where $g_1(\boldsymbol{X},U)=1-X_1^2+X_2^2+\xi X_1X_2U$; $g_2(\boldsymbol{X},U,a)=a(4+X_1-0.25\xi X_1^2-2\times 1\{X_2\leqslant 0\})$; then we generate treatment assignment following

$$\log\left\{\frac{P(A=1\mid \boldsymbol{X},U,Z)}{P(A=-1\mid \boldsymbol{X},U,Z)}\right\}=8Z+X_1-7X_2.$$

In this setting, treatment assignment depends on X and Z, but not on the unmeasured variable U,

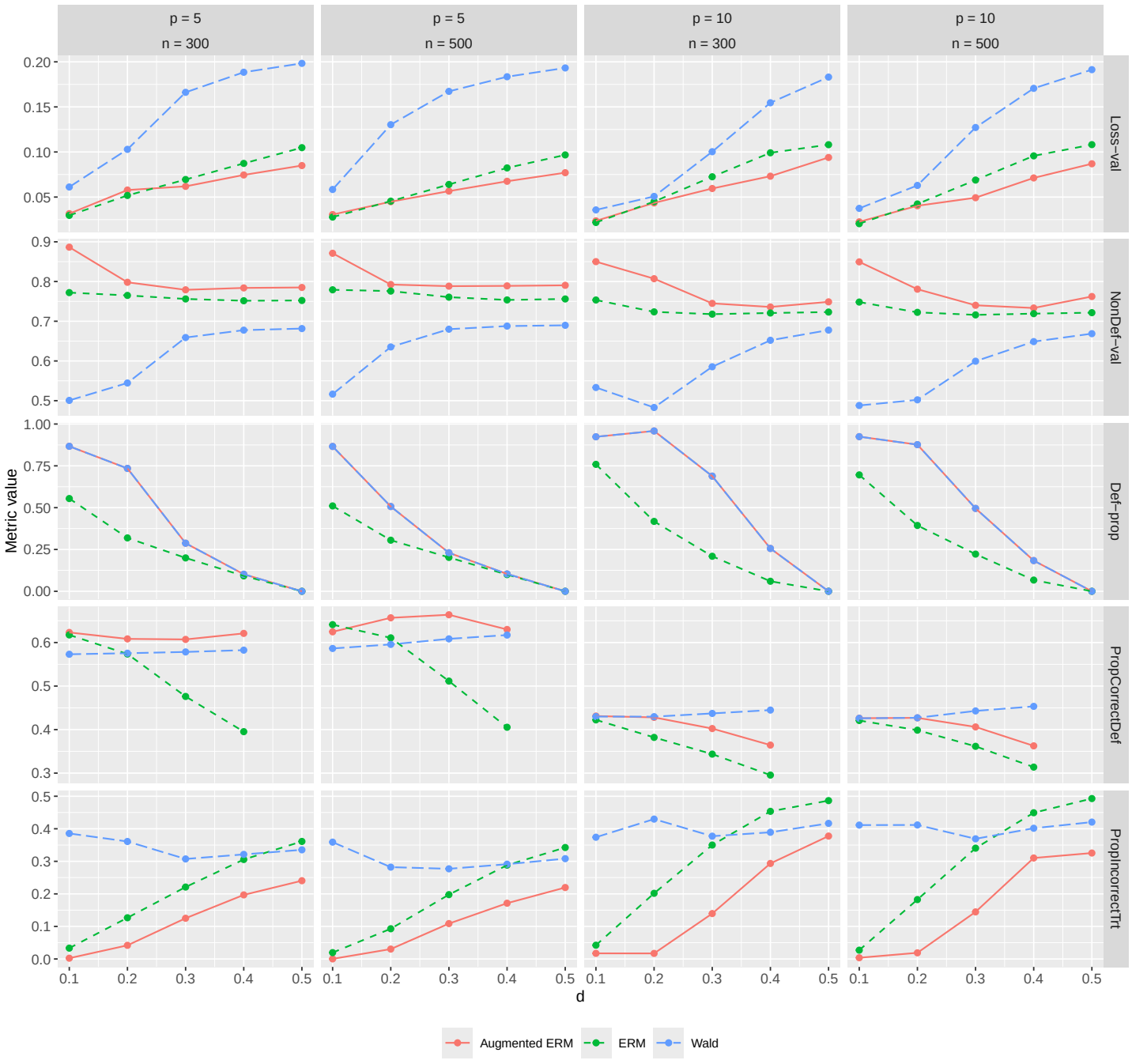


Figure 7: Simulation results under varying dimensions over 200 repeats.

i.e., $A \perp U \mid X, Z$. Thus, the CATE is identifiable and can be consistently estimated using the Wald estimator. In our implementation of the Wald estimator, we estimate $\Delta_Y(\boldsymbol{X})$ and $\Delta_A(\boldsymbol{X})$ separately using kernel regressions, and then estimate the CATEs; we do not implement any deferred option for the Wald estimator since in this setting, the CATE is identifiable. We report the worst total loss in value and the proportion of incorrect treatment decisions. For the proportion of incorrect treatment decisions, we only calculate based on the patients not receiving the deferred option. Notice that our proposed methods are implemented as if we did not know that the CATE is identifiable via the Wald estimator; otherwise, we can replace L and U to be any consistent estimator of CATEs, and we expect similar performance to using the Wald estimator directly.

Figure 8 summarizes the results. When $d = 0.5$, for which no deferral is allowed, the Wald estimator achieves the lowest worst-case total loss and the lowest proportion of incorrect treatment assignments. This is expected because the CATE is point identified and the Wald estimator is consistent in this setting. When $d < 0.5$, the proposed methods allow deferral and achieve lower worst-case total loss, as well as fewer incorrect treatment decisions among patients receiving a definitive recommendation. In this case, the proportion of deferred decisions increases as d decreases, illustrating the trade-off controlled by d. In summary, when point identification is known to hold, the point-identification approach is preferable if a treatment recommendation is required for every patient. The proposed framework instead provides a mechanism for trading off definitive recommendations against uncertainty when one does not wish to rely on the stronger point-identification assumptions.

5 Additional results for application

In this section, we provide additional results for the real data analysis in the main text. In the main text, to examine the estimated ITRs under different risks of taking the deferred option, we vary d from 1 to 0.11. We observe that with the decrease of d, more and more patients would take the deferred option indicating that they are affected by the non-adherence issue and their optimal decisions are

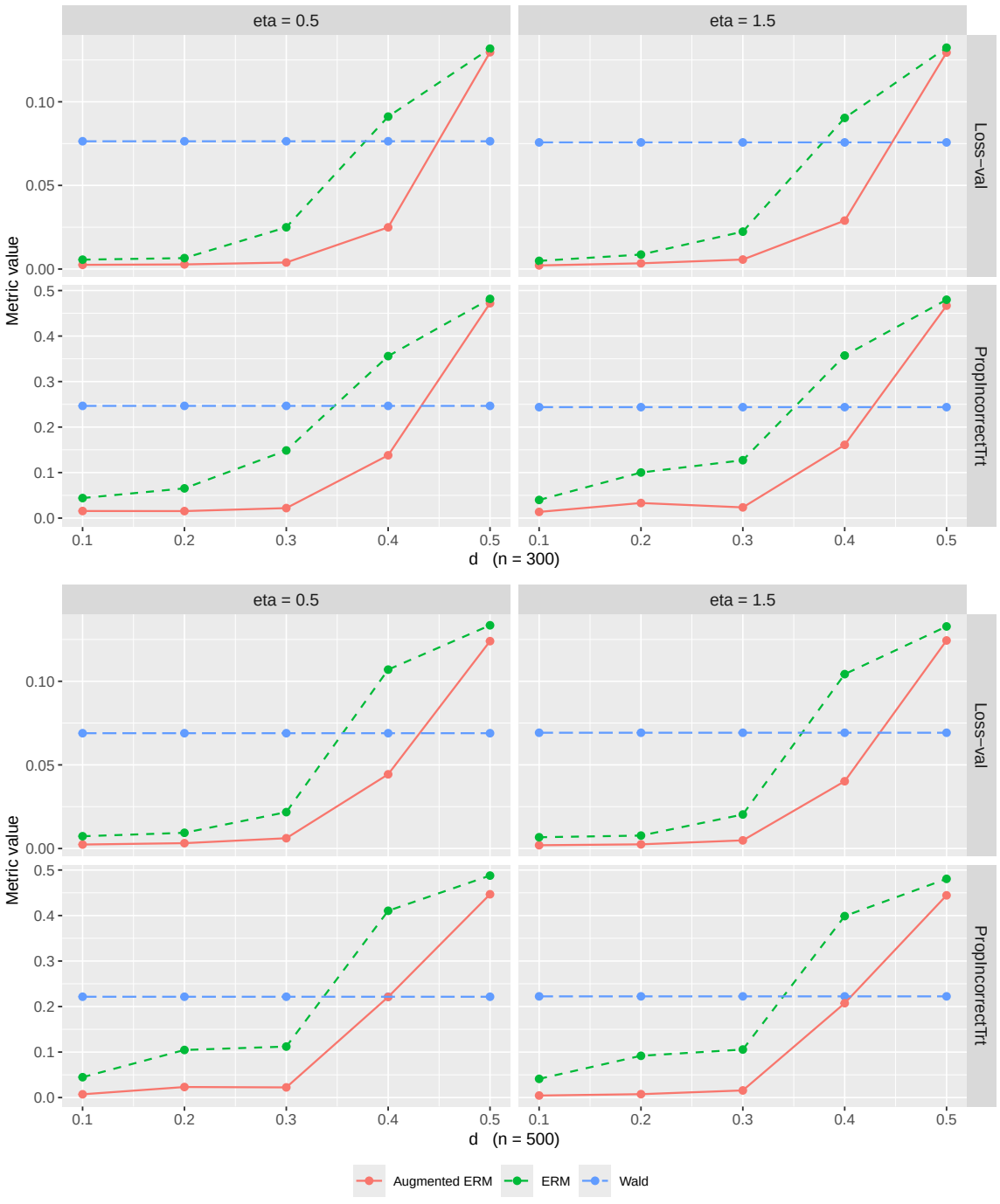


Figure 8: Simulation results under varying confounding strength over 200 repeats.

susceptible to the identification uncertainty. In this analysis, we consider the treatment assignment for those who are suggested to take deferred option by the estimated ITRs with the deferred option. For these patients, we will further suggest to take finasteride if there is a reduced risk of developing all grades of prostate cancer. We use the initial randomized treatment and a binary outcome indicating whether any prostate cancer occurs to estimate an ITR by the proposed method in Zhao et al. (2012). Then for those who are suggested to take the deferred option by our proposed method, we follow the estimated ITR by the method in Zhao et al. (2012) to make their treatment decisions. Figure 9 illustrates treatment decisions for all the patients. We can see that there is a relatively stable subgroup for whom we suggest against finasteride; by using the estimated ITR to reduce overall risk of developing prostate cancer, majority of those who take the deferred option will be suggested to take finasteride. This indicates that there might be a subgroup who can safely benefit from the use of finasteride. A further study based on a randomized clinical trial might be needed to ascertain their potential benefits.

In the following analysis, we conduct a sensitivity analysis regarding the choice of the adherence threshold. In the main text, the 80% cutoff was not chosen arbitrarily but but follows established practice in existing literature Pauler et al. (2002); Pinsky et al. (2021). To evaluate the robustness, we further conduct a sensitivity analysis, where the threshold was changed to 90%. Figure 10 and 11 summarized the results with varying thresholds. We can see that the conclusion of the real data analysis remains similar, i.e., most of patients will be affected by the non-adherence issue and thus their optimal treatment decisions cannot be identified.

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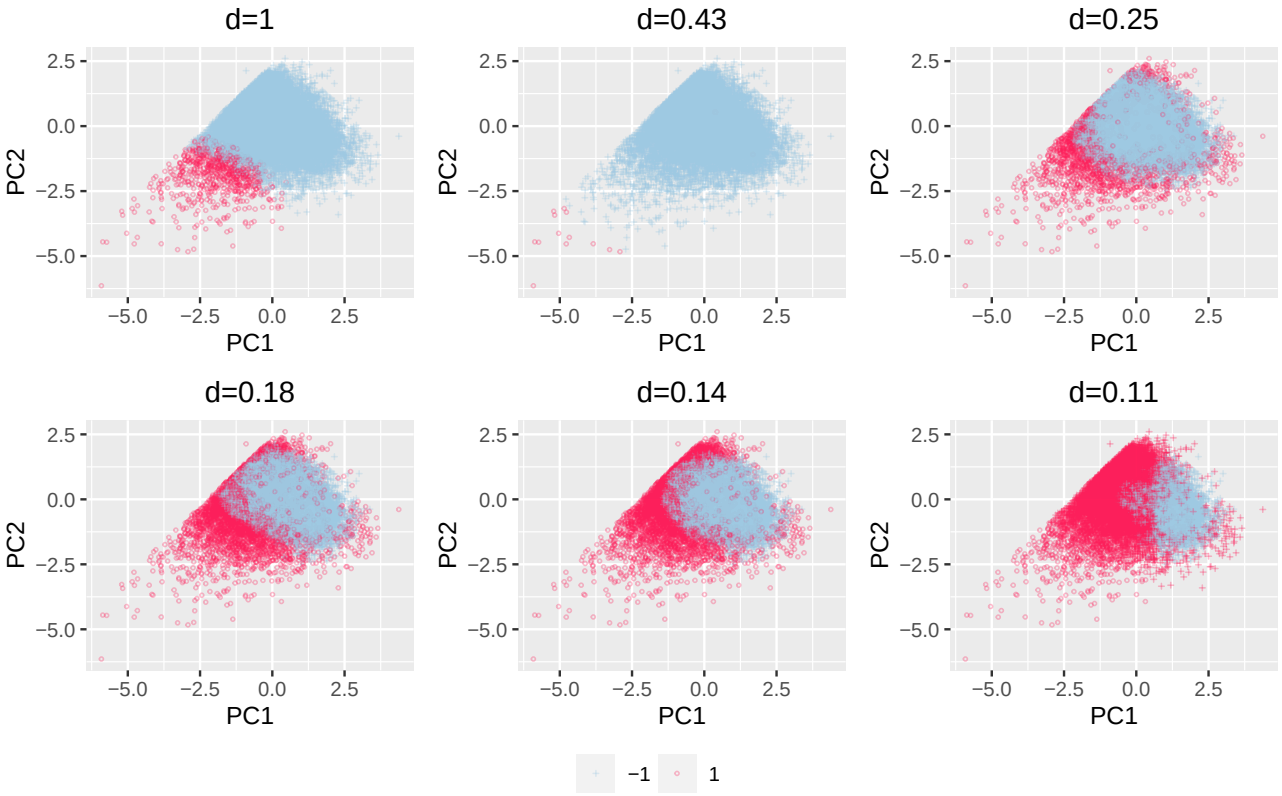


Figure 9: Decisions of the estimated ITRs with varying discount factor d.

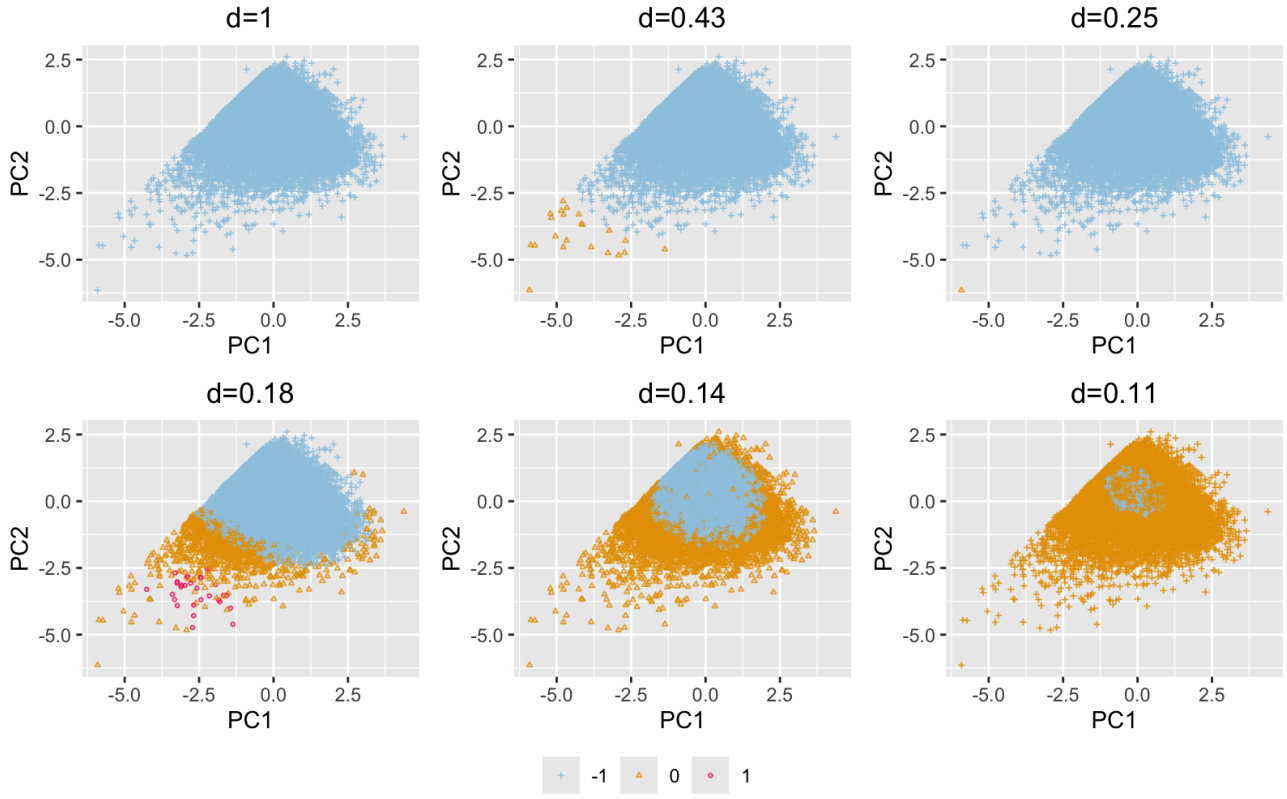


Figure 10: Decisions of the estimated ITRs under 90% adherence threshold.

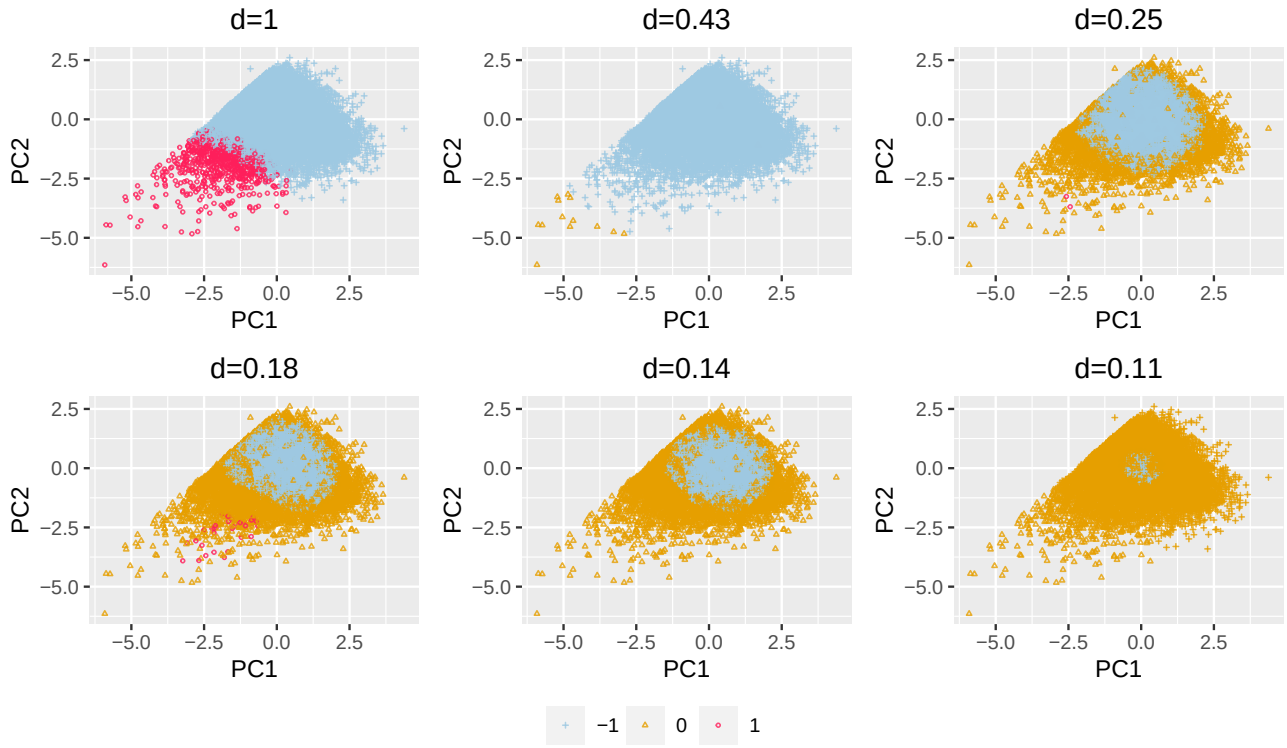


Figure 11: Decisions of the estimated ITRs under 80% adherence threshold.

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